

Towards Trustworthy Causal Artificial Intelligence for Predictive Maintenance: A Structural Causal Analysis of NASA C-MAPSS Turbofan Engine Degradation

A Technical Research Report

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Keywords

Trustworthy Artificial Intelligence, Causal Artificial Intelligence, Predictive Maintenance, Remaining Useful Life Prediction, NASA C-MAPSS, Industrial Artificial Intelligence, Structural Causal Models, Temporal Causal Discovery, Counterfactual Reasoning, Structural Equation Modelling, Machine Learning, Digital Maintenance, Explainable Artificial Intelligence.

Abstract

Predictive maintenance has emerged as one of the most influential applications of artificial intelligence within modern industrial systems, enabling organisations to anticipate equipment degradation before catastrophic failures occur. Accurate estimation of Remaining Useful Life (RUL) reduces operational costs, improves asset reliability, and enables condition-based maintenance scheduling. Despite remarkable advances in machine learning, most predictive maintenance systems remain fundamentally correlational, limiting their ability to explain degradation mechanisms or support intervention-oriented decision making.

This technical report presents the design and implementation of a Trustworthy Causal Artificial Intelligence framework for predictive maintenance using the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) turbofan engine degradation dataset. The proposed framework extends traditional predictive modelling by integrating Remaining Useful Life estimation, explainable artificial intelligence, temporal causal discovery, structural causal modelling, structural equation modelling, counterfactual reasoning, uncertainty estimation, and intervention-based maintenance optimisation into a unified analytical pipeline.

The research was implemented through eight progressively developed computational notebooks. The initial stages focused on exploratory data analysis, Remaining Useful Life engineering, and predictive modelling using Random Forest regression. Explainable artificial intelligence techniques were subsequently incorporated to identify degradation-sensitive sensors contributing to predictive performance. Building upon these findings, temporal dependencies among engine sensors were investigated using Granger causality analysis, enabling the construction of a directed causal dependency network representing degradation propagation throughout the engine system.

The discovered temporal relationships were formalised into a Structural Causal Model (SCM) and further quantified using Structural Equation Modelling (SEM), allowing both direct and indirect causal effects to be estimated. Counterfactual reasoning was then employed to investigate hypothetical maintenance scenarios by estimating changes in Remaining Useful

Life following simulated sensor interventions. Finally, a maintenance intervention optimisation framework was developed to prioritise corrective actions according to estimated causal influence and intervention effectiveness, providing practical decision support for industrial maintenance planning.

Experimental results demonstrate that the proposed framework successfully integrates predictive modelling, explainable artificial intelligence, temporal causal discovery, structural causal modelling, and counterfactual reasoning into a unified decision-support pipeline. Unlike conventional predictive maintenance approaches that focus primarily on Remaining Useful Life estimation, the proposed framework enables interpretable degradation analysis, causal intervention assessment, and maintenance prioritisation. The work demonstrates the feasibility of transitioning predictive maintenance from correlation-based forecasting towards trustworthy causal decision intelligence for future industrial applications.

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Chapter 1

Introduction

1.1 Background

Industrial equipment forms the foundation of modern manufacturing, transportation, aerospace, energy production, and large-scale infrastructure systems. The continuous operation of these assets is essential for maintaining productivity, ensuring operational safety, and reducing economic losses associated with unexpected equipment failures. Traditionally, industrial maintenance strategies have relied upon either corrective maintenance, where repairs are performed only after failure has occurred, or preventive maintenance, where components are replaced according to predetermined maintenance schedules regardless of their actual condition. While corrective maintenance frequently results in costly operational interruptions and potential safety hazards, preventive maintenance often leads to unnecessary replacement of components that remain operational, increasing maintenance expenditure without fully utilising component lifetimes.

The emergence of Industry 4.0 has transformed industrial maintenance by introducing intelligent monitoring systems capable of continuously analysing sensor data generated by complex engineering assets. These developments have enabled the widespread adoption of predictive maintenance, an advanced maintenance strategy that estimates equipment health using historical operational data and machine learning algorithms. Rather than relying solely upon fixed maintenance intervals, predictive maintenance seeks to identify degradation trends and forecast future failures before they occur, allowing maintenance activities to be scheduled proactively according to the actual condition of the equipment.

Among predictive maintenance tasks, Remaining Useful Life (RUL) prediction has received significant research attention. Remaining Useful Life represents the estimated number of operational cycles remaining before an asset reaches functional

failure. Accurate RUL estimation allows maintenance engineers to minimise unexpected downtime while simultaneously avoiding unnecessary preventive maintenance. Consequently, RUL prediction has become one of the most important applications of artificial intelligence within industrial asset management.

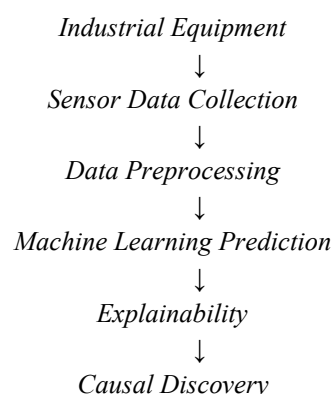
Recent advances in machine learning have significantly improved predictive performance for Remaining Useful Life estimation. Algorithms including Random Forests, Support Vector Machines, Gradient Boosting methods, Artificial Neural Networks, Long Short-Term Memory (LSTM) networks, and Transformer-based architectures have demonstrated considerable success across numerous industrial benchmark datasets. These methods primarily exploit statistical relationships between historical sensor measurements and degradation targets to generate accurate predictions.

However, despite their predictive capability, conventional machine learning models remain fundamentally correlational. Their internal representations frequently provide limited insight into the mechanisms responsible for equipment degradation. Consequently, while such models may accurately predict that an engine is approaching failure, they often fail to answer questions that are of central importance to maintenance engineers, including why degradation occurred, which sensor interactions contributed to failure progression, and how maintenance interventions might alter the degradation process. This limitation has motivated increasing interest in Trustworthy Artificial Intelligence, a research area concerned with developing machine learning systems that provide transparency, interpretability, robustness, uncertainty awareness, and human-understandable reasoning. Within industrial environments, trustworthy AI is particularly important because maintenance decisions frequently involve high economic costs and safety-critical consequences. Decision-support systems must therefore provide explanations that extend beyond prediction accuracy to include causal understanding of degradation mechanisms.

One promising research direction involves the integration of Causal Artificial Intelligence into predictive maintenance systems. Unlike conventional machine learning approaches that identify statistical associations, causal artificial intelligence seeks to identify directional relationships capable of explaining how degradation propagates throughout complex engineering systems. Causal reasoning further enables counterfactual analysis, allowing engineers to investigate hypothetical maintenance interventions and estimate their potential impact on equipment health before implementation.

The NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset has become one of the most widely adopted benchmark datasets for Remaining Useful Life prediction. The dataset provides multivariate time-series measurements collected from simulated turbofan engines operating under progressive degradation conditions. Its realistic degradation trajectories, multiple operational settings, and extensive sensor measurements make it an ideal benchmark for evaluating predictive maintenance methodologies while simultaneously supporting research into causal degradation analysis.

This technical report presents the development of a Trustworthy Causal Artificial Intelligence framework that extends conventional predictive maintenance by integrating predictive modelling, explainable artificial intelligence, temporal causal discovery, structural causal modelling, counterfactual reasoning, and intervention optimisation within a unified analytical workflow. The objective is not merely to improve predictive accuracy, but to provide interpretable and intervention-aware decision support capable of advancing industrial artificial intelligence toward trustworthy and causally informed predictive maintenance systems.



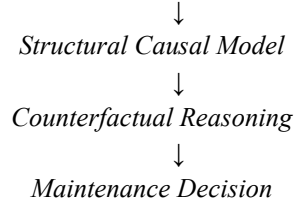


Figure 1.1: Industrial Predictive Maintenance Workflow

1.2 Problem Statement

Despite significant advances in predictive maintenance, the majority of industrial artificial intelligence systems remain fundamentally prediction-oriented rather than reasoning-oriented. Existing machine learning models are generally optimised to maximise predictive accuracy by learning statistical relationships between historical sensor observations and equipment failure indicators. Although these approaches often achieve competitive performance on benchmark datasets, they provide limited understanding of the underlying degradation mechanisms responsible for equipment failure.

This limitation presents a considerable challenge for industrial decision-making. Maintenance engineers rarely require predictions alone; instead, they require explanations that support operational decisions. Practical maintenance planning requires answers to questions such as which subsystem-initiated degradation, how deterioration propagated through interconnected components, and which maintenance intervention is most likely to improve equipment health. Conventional predictive models are generally unable to provide such explanations because they identify statistical associations rather than causal relationships.

Furthermore, many explainable artificial intelligence techniques introduced for predictive maintenance remain restricted to feature attribution. Methods such as feature importance ranking, SHAP values, and permutation importance identify variables that contribute strongly to model predictions. However, these approaches cannot distinguish between predictive correlation and genuine causal influence. A sensor receiving a high importance score may simply respond to degradation occurring elsewhere within the system rather than acting as the initiating cause of failure.

Another significant limitation concerns intervention planning. Industrial maintenance ultimately seeks to improve equipment health rather than simply predict failure. Correlation-based models cannot estimate the consequences of hypothetical maintenance actions because they do not represent causal mechanisms governing system behaviour. Consequently, existing predictive maintenance systems frequently lack the ability to support counterfactual reasoning or intervention-based optimisation.

The challenge becomes even more important when predictive models are deployed within safety-critical industrial environments. Decision-support systems that influence maintenance scheduling, operational planning, and asset management must provide reliable explanations that human operators can interpret, verify, and trust. Increasing regulatory emphasis on trustworthy artificial intelligence further highlights the importance of transparency, robustness, uncertainty estimation, and explainability in industrial machine learning applications.

Although causal artificial intelligence has recently emerged as a promising research direction, its application within predictive maintenance remains comparatively limited. Most existing industrial studies focus exclusively on predictive performance, while relatively few attempt to discover temporal causal relationships between degradation variables, construct structural causal models, or perform counterfactual maintenance analysis. Consequently, there remains a significant research gap between accurate predictive modelling and trustworthy causal decision-support systems.

This project addresses that gap by extending conventional Remaining Useful Life prediction into a comprehensive causal artificial intelligence framework capable of integrating prediction, explanation, temporal causal discovery, structural causal

modelling, counterfactual reasoning, uncertainty estimation, and maintenance intervention optimisation within a unified workflow.

1.3 Research Motivation

The motivation for this research originates from the observation that industrial artificial intelligence has progressed considerably in predictive capability while remaining comparatively limited in causal understanding. Modern machine learning algorithms can identify complex degradation patterns from high-dimensional sensor measurements; however, these models frequently behave as statistical function approximators whose internal reasoning remains difficult to interpret.

Within industrial environments, predictive accuracy alone is insufficient for supporting maintenance decisions. Maintenance engineers require actionable knowledge regarding degradation progression rather than numerical predictions in isolation. An artificial intelligence system capable of estimating Remaining Useful Life should ideally explain why degradation is occurring, identify the variables responsible for failure progression, quantify confidence in its predictions, and estimate the potential consequences of alternative maintenance strategies.

This research therefore seeks to bridge the gap between predictive machine learning and causal decision intelligence. Rather than treating prediction as the final objective, the project considers prediction as the first stage of a broader reasoning framework. Explainability, causal discovery, structural causal modelling, and intervention analysis are progressively incorporated to enable more informative maintenance recommendations.

The NASA C-MAPSS turbofan degradation dataset provides an appropriate experimental environment for investigating these concepts because it contains multivariate degradation trajectories representative of realistic industrial systems. The dataset enables predictive modelling while simultaneously supporting investigation of temporal dependencies among degradation-sensitive sensors.

An additional motivation arises from the increasing importance of trustworthy artificial intelligence within industrial applications. Contemporary AI systems are expected not only to achieve strong predictive performance but also to provide transparency, robustness, interpretability, fairness, and uncertainty awareness. The integration of causal reasoning represents an important step towards achieving these objectives by enabling models to distinguish between statistical association and genuine cause-and-effect relationships.

Finally, this research reflects the broader transition currently occurring within artificial intelligence research, where increasing emphasis is placed upon intervention-aware machine learning capable of supporting scientific discovery, engineering analysis, and evidence-based decision making. By integrating causal artificial intelligence with predictive maintenance, this work contributes toward developing industrial AI systems that support both accurate prediction and trustworthy reasoning.

1.4 Research Objectives

The primary objective of this research is to develop a Trustworthy Causal Artificial Intelligence framework capable of extending conventional Remaining Useful Life prediction into an interpretable and intervention-aware predictive maintenance system.

To achieve this objective, the following research goals were established:

- Develop an accurate Remaining Useful Life prediction model using the NASA C-MAPSS turbofan engine degradation dataset.
- Investigate degradation-sensitive sensor behaviour through exploratory analysis and feature engineering.

- Integrate explainable artificial intelligence techniques to identify influential variables contributing to predictive performance.
- Discover temporal causal relationships among engine sensors using Granger causality analysis.
- Construct a directed Structural Causal Model representing degradation propagation throughout the engine system.
- Quantify causal dependencies using Structural Equation Modelling.
- Investigate hypothetical maintenance scenarios through counterfactual reasoning.
- Estimate predictive uncertainty to improve model trustworthiness.
- Develop an intervention prioritisation framework capable of recommending maintenance actions according to estimated causal influence.
- Demonstrate how predictive maintenance systems can transition from statistical prediction towards trustworthy causal decision-support.

1.5 Research Contributions

This research contributes to the field of industrial artificial intelligence by proposing an integrated framework that combines predictive modelling with causal reasoning for trustworthy predictive maintenance. Unlike conventional Remaining Useful Life prediction studies that terminate after model evaluation, the proposed methodology extends the analytical pipeline toward explanation, causal discovery, structural modelling, and intervention analysis.

The principal contributions of this work are summarised as follows. First, a complete Remaining Useful Life prediction pipeline was developed using the NASA C-MAPSS turbofan engine degradation dataset. The framework includes data preprocessing, Remaining Useful Life engineering, feature standardisation, model development, and predictive evaluation using a Random Forest regression model.

Second, explainable artificial intelligence techniques were incorporated to investigate sensor importance and degradation behaviour. Rather than treating the predictive model as a black box, feature importance analysis was employed to identify the most influential degradation-related sensors contributing to Remaining Useful Life estimation.

Third, temporal causal discovery was introduced through Granger causality analysis to investigate directional dependencies between degradation-sensitive sensors. This stage represents an important transition from purely statistical association toward temporal causal reasoning.

Fourth, the temporal relationships identified through causal discovery were formalised into a Structural Causal Model (SCM), providing an explicit graphical representation of degradation propagation within the turbofan engine system. Structural Equation Modelling (SEM) was subsequently employed to quantify direct and indirect causal effects among degradation variables.

Fifth, counterfactual reasoning was incorporated to simulate hypothetical maintenance interventions. Instead of merely predicting engine failure, the proposed framework estimates how Remaining Useful Life would change under alternative maintenance scenarios, thereby supporting intervention-aware decision making.

Sixth, prediction uncertainty and intervention prioritisation were integrated into the analytical pipeline to improve system trustworthiness. Maintenance actions were ranked according to estimated causal influence, enabling engineers to prioritise corrective actions based upon expected operational benefit rather than predictive correlation alone.

Finally, this research demonstrates how predictive machine learning, explainable artificial intelligence, and causal inference can be integrated into a unified framework for trustworthy industrial decision support.

1.6 Scope of the Study

The scope of this research is intentionally focused on developing and evaluating a causal artificial intelligence framework using a publicly available benchmark dataset for predictive maintenance research.

The experimental investigation is conducted exclusively using the FD001 subset of the NASA C-MAPSS turbofan engine degradation dataset. This subset represents a single operating condition and a single failure mode, allowing degradation behaviour to be analysed under relatively controlled operating circumstances.

The predictive component of the framework focuses on Remaining Useful Life estimation using supervised machine learning techniques. Random Forest regression was selected as the primary predictive model because of its robustness, interpretability, and suitability for tabular industrial sensor data. The objective of the predictive stage is not necessarily to achieve state-of-the-art predictive accuracy but rather to provide a reliable foundation upon which causal reasoning can subsequently be developed.

Temporal causal discovery is performed using Granger causality analysis applied to the degradation-sensitive sensors identified during the explainability stage. The resulting temporal dependency network is subsequently transformed into a Structural Causal Model representing hypothesised degradation pathways within the engine system.

Structural Equation Modelling is employed to quantify the strength of causal relationships identified during temporal analysis. Counterfactual reasoning is then used to simulate hypothetical maintenance interventions by estimating the effects of modifying selected degradation variables.

Although the proposed framework demonstrates a complete causal reasoning pipeline, several limitations remain. The Structural Causal Model is constructed from observational data rather than controlled experimental interventions and therefore represents an approximate causal representation rather than definitive physical causation. Furthermore, the analysis is limited to a single benchmark dataset and does not incorporate heterogeneous industrial operating environments, multimodal sensor fusion, or real-world maintenance records.

Consequently, the proposed framework should be regarded as a research-oriented proof of concept demonstrating the integration of predictive modelling with causal artificial intelligence rather than a fully deployed industrial maintenance system.

1.7 Report Organization

The remainder of this technical report is organised as follows.

Chapter 2 reviews the current state of research in predictive maintenance, Remaining Useful Life prediction, explainable artificial intelligence, causal machine learning, structural causal modelling, and trustworthy artificial intelligence. The chapter identifies existing research limitations and establishes the motivation for the proposed framework.

Chapter 3 introduces the NASA C-MAPSS dataset used throughout the study, including its operational characteristics, sensor measurements, Remaining Useful Life formulation, and preprocessing methodology.

Chapter 4 describes the overall methodological framework developed in this research. The predictive modelling pipeline, explainability techniques, temporal causal discovery, structural causal modelling, structural equation modelling, counterfactual reasoning, uncertainty estimation, and intervention prioritisation procedures are discussed in detail.

Chapter 5 presents the experimental results obtained during each stage of the analytical pipeline, including predictive performance, explainability analysis, temporal causal relationships, structural causal modelling results, and maintenance intervention simulations.

Chapter 6 discusses the implications of the proposed framework for trustworthy industrial artificial intelligence. The advantages, limitations, and practical significance of integrating causal reasoning into predictive maintenance are critically analysed.

Chapter 7 summarises the principal findings of this research and outlines potential directions for future work involving digital twins, probabilistic causal inference, reinforcement learning, and intervention-aware industrial artificial intelligence systems.

Chapter 2

Literature Review

2.1 Introduction

The rapid adoption of Industry 4.0 technologies has transformed modern industrial maintenance by enabling continuous monitoring of equipment through large-scale sensor networks, cyber-physical systems, and intelligent data analytics. As industrial systems become increasingly connected, predictive maintenance has emerged as one of the most significant applications of artificial intelligence, offering substantial improvements in operational reliability, maintenance efficiency, and asset utilisation.

Predictive maintenance aims to estimate equipment health before catastrophic failure occurs by analysing historical operational data. Among the various predictive maintenance tasks, Remaining Useful Life (RUL) prediction has received particular attention because it provides maintenance engineers with quantitative estimates of the operational lifetime remaining before functional failure. Accurate Remaining Useful Life estimation enables maintenance scheduling based on actual equipment condition rather than predetermined maintenance intervals, reducing both unexpected failures and unnecessary preventive maintenance. Although significant advances have been achieved through machine learning and deep learning methods, contemporary predictive maintenance systems remain primarily prediction-oriented. Their internal decision-making processes often remain difficult to interpret, limiting their practical usefulness in safety-critical industrial environments. Recent research therefore increasingly emphasises explainability, uncertainty estimation, and causal reasoning as essential components of trustworthy industrial artificial intelligence.

This chapter reviews the existing literature relevant to the present study. Attention is given to Remaining Useful Life prediction, explainable artificial intelligence, causal machine learning, structural causal modelling, and trustworthy artificial intelligence. The chapter concludes by identifying the research gap addressed by the proposed framework.

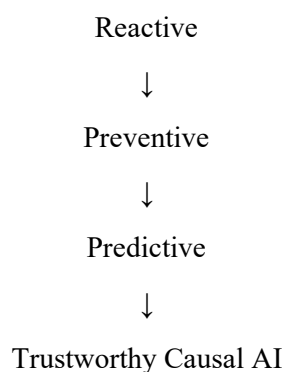


Figure 2.1: Evolution of maintenance Strategies

2.2 Predictive Maintenance

Predictive maintenance has become one of the most important applications of artificial intelligence within modern industrial systems. Unlike reactive maintenance, where corrective actions are performed only after equipment failure, or preventive maintenance, where maintenance activities are scheduled at fixed intervals regardless of equipment condition, predictive maintenance continuously monitors equipment health in order to estimate future degradation and optimise maintenance planning.

The emergence of Industry 4.0 has significantly accelerated the development of predictive maintenance systems through the widespread deployment of industrial sensors, Internet of Things (IoT) devices, cyber-physical systems, cloud computing infrastructures, and advanced machine learning algorithms. Modern industrial assets continuously generate large volumes of operational data, enabling artificial intelligence models to identify subtle degradation patterns that would otherwise remain undetected by conventional inspection procedures.

The primary objective of predictive maintenance is to maximise equipment availability while simultaneously reducing maintenance cost, unexpected downtime, and operational risk. By anticipating equipment degradation before catastrophic failure occurs, maintenance activities can be scheduled during planned operational windows, improving both production efficiency and asset utilisation.

Typical predictive maintenance systems consist of several sequential stages. Initially, raw sensor measurements are acquired from industrial equipment through monitoring systems. These measurements undergo preprocessing procedures including data cleaning, normalization, feature engineering, and temporal aggregation. Machine learning models subsequently analyse the processed data to estimate equipment health, detect anomalies, or predict Remaining Useful Life. Finally, maintenance recommendations are generated to support operational decision-making.

Predictive maintenance has been successfully applied across numerous industrial domains including aerospace, manufacturing, transportation, energy production, offshore engineering, railway systems, and healthcare equipment management. Despite differences in application domains, most predictive maintenance frameworks follow similar analytical pipelines involving sensor data acquisition, degradation modelling, predictive analytics, and maintenance optimisation.

Although predictive maintenance has achieved substantial practical success, many existing systems remain largely focused on prediction accuracy while providing limited explanation regarding the mechanisms responsible for equipment degradation. Consequently, recent research increasingly investigates methods capable of combining predictive modelling with interpretable and trustworthy reasoning.

2.3 Remaining Useful Life Prediction

Remaining Useful Life (RUL) prediction represents one of the most extensively studied problems within predictive maintenance research. Remaining Useful Life refers to the estimated operational duration that a machine or component can continue functioning before reaching an unacceptable degradation threshold or complete failure.

Formally, Remaining Useful Life may be defined as

$$RUL = T_f - T_c$$

where T_f denotes the failure time and T_c represents the current operating time.

Unlike binary fault detection, Remaining Useful Life estimation provides continuous degradation information, enabling maintenance engineers to optimise inspection schedules, spare-part logistics, maintenance planning, and operational risk

assessment. Consequently, Remaining Useful Life prediction has become particularly important within industries operating high-value assets such as aircraft engines, gas turbines, wind turbines, nuclear facilities, and railway infrastructure.

Traditional Remaining Useful Life estimation methods relied primarily upon physics-based degradation models describing failure mechanisms using engineering knowledge. While these approaches often provide strong physical interpretability, they generally require extensive domain expertise and may not generalise effectively across different operating environments.

Data-driven machine learning approaches have therefore become increasingly popular due to their ability to automatically learn degradation behaviour directly from historical sensor measurements. These methods require fewer assumptions regarding underlying physical processes and can often capture highly nonlinear relationships between sensor behaviour and equipment degradation.

The NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset has become one of the most widely used benchmark datasets for Remaining Useful Life prediction. The dataset simulates progressive degradation of turbofan aircraft engines operating under controlled conditions and provides multivariate sensor measurements collected throughout the engine lifecycle. Because the true Remaining Useful Life is known for every engine trajectory, the dataset enables objective evaluation and comparison of predictive maintenance algorithms.

Recent studies have employed a wide range of machine learning techniques for Remaining Useful Life prediction, including Random Forest regression, Support Vector Regression, Artificial Neural Networks, Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), Transformer architectures, Graph Neural Networks, and hybrid deep learning frameworks. Although these methods often achieve impressive predictive performance, they generally provide limited explanation regarding why particular degradation trajectories produce specific Remaining Useful Life estimates.

2.4 Machine Learning for Predictive Maintenance

Machine learning has fundamentally transformed predictive maintenance by enabling automated extraction of degradation patterns from complex industrial sensor data. Rather than relying upon manually designed degradation rules, machine learning algorithms identify statistical relationships directly from historical observations.

Supervised learning represents the dominant machine learning paradigm for predictive maintenance because labelled degradation data are frequently available from historical maintenance records or simulation environments such as NASA C-MAPSS. Under supervised learning, models learn mappings between sensor observations and target variables including Remaining Useful Life, fault category, or health state.

Among classical machine learning methods, Random Forest regression has received considerable attention because of its robustness, resistance to overfitting, ability to model nonlinear relationships, and relatively straightforward interpretability. Random Forest models construct ensembles of decision trees trained using bootstrap aggregation and random feature selection, thereby reducing prediction variance while maintaining strong generalisation performance.

Support Vector Machines and Gradient Boosting algorithms have also demonstrated competitive performance across numerous predictive maintenance datasets. More recently, deep learning approaches have become increasingly dominant due to their capacity to automatically learn hierarchical feature representations from raw sequential sensor measurements.

Long Short-Term Memory (LSTM) networks have become particularly popular because they effectively model temporal dependencies present within degradation trajectories. Convolutional Neural Networks have similarly demonstrated success by automatically extracting local degradation features from multivariate sensor sequences. More recently, Transformer architectures have attracted significant interest because of their ability to model long-range temporal dependencies using self-attention mechanisms.

Despite these advances, high predictive accuracy does not necessarily imply trustworthy decision-making. Deep neural networks frequently behave as highly complex black-box models whose internal reasoning remains difficult to interpret. Consequently, increasing attention has shifted toward developing machine learning methods capable of providing transparent explanations, uncertainty estimates, and causal reasoning alongside predictive performance.

The integration of explainability and causal inference therefore represents one of the most active research directions within trustworthy industrial artificial intelligence and forms the central motivation for the framework proposed in this report.

2.5 Causal Discovery Methodology

Conventional predictive maintenance models identify statistical relationships between degradation variables; however, statistical dependence alone cannot establish directional influence between engine components. In industrial systems, understanding how degradation propagates through interconnected subsystems is essential for reliable maintenance planning and trustworthy decision support.

To address this limitation, temporal causal discovery was incorporated into the proposed framework. Instead of analysing sensor relationships solely through correlation coefficients, this stage investigates whether historical observations from one sensor improve the prediction of another sensor over time. Such temporal precedence provides stronger evidence for potential causal influence than static association alone.

The causal discovery process consisted of four sequential stages. First, sensor measurements were standardised using z-score normalisation to remove differences in physical measurement scales while preserving degradation dynamics.

Second, each sensor time series was evaluated using the Augmented Dickey–Fuller (ADF) statistical test. Stationarity is a prerequisite for reliable Granger causality analysis because non-stationary signals may produce misleading causal relationships caused by long-term trends rather than genuine temporal dependence.

Third, Granger causality tests were performed between the degradation-sensitive sensors identified during the explainability stage. For every possible sensor pair, the null hypothesis assumes that the historical values of one sensor do not improve prediction of the other. Rejection of this hypothesis indicates the existence of statistically significant temporal influence.

Finally, statistically significant causal relationships were represented as a directed network, where each node corresponds to an engine sensor and each directed edge represents a significant temporal dependency. Unlike traditional feature importance analysis, this network represents potential pathways through which degradation propagates across engine components.

2.6 Results of Temporal Causal Discovery

The Augmented Dickey–Fuller analysis demonstrated that all selected degradation sensors satisfied the stationarity assumption required for Granger causality analysis. Extremely small p-values were obtained for every investigated variable, indicating that the null hypothesis of non-stationarity could be rejected with high confidence.

This result suggests that the observed temporal variations originate primarily from degradation behaviour rather than uncontrolled statistical drift, providing an appropriate basis for subsequent causal modelling. Following stationarity verification, pairwise Granger causality tests were conducted among the selected degradation sensors.

Only statistically significant relationships satisfying the predefined significance threshold ($p < 0.05$) were retained for further analysis. The resulting causal graph contained multiple directional dependencies, revealing that degradation does not occur independently across sensors but instead propagates through interconnected measurement variables. Rather than

identifying isolated sensor importance, the causal graph illustrates a network of interacting degradation mechanisms that collectively influence engine health.

Several sensors appeared repeatedly as dominant causal drivers, indicating that changes within these measurements precede degradation observed in other parts of the engine. These variables therefore become suitable candidates for subsequent intervention analysis and structural causal modelling.

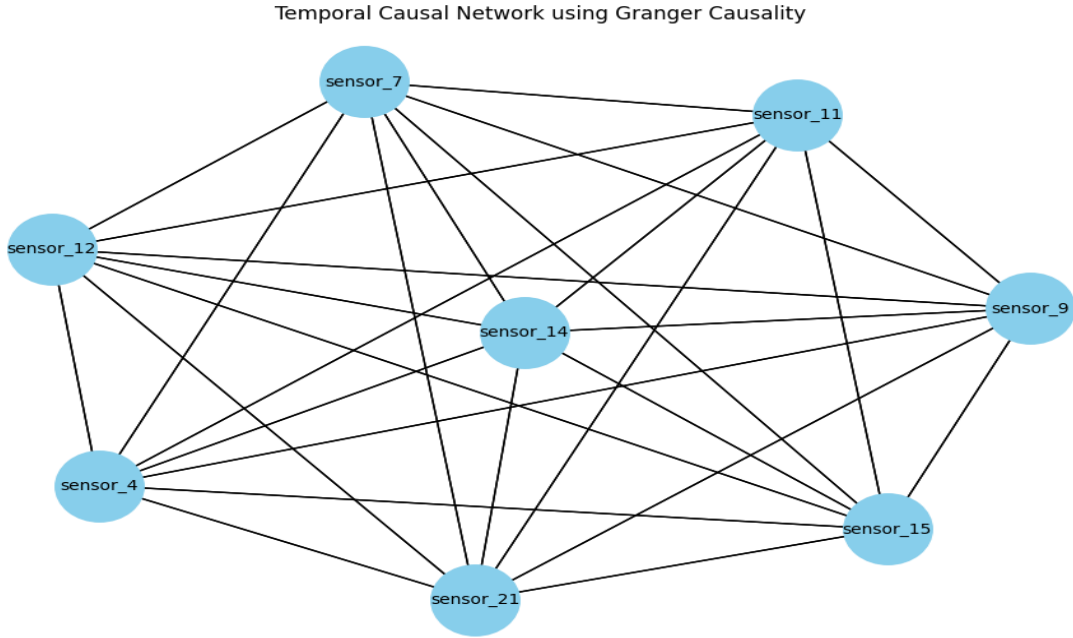


Figure 2.2: Temporal Granger Causal Network among Selected Degradation Sensors

2.7 Interpretation of the Temporal Causal Network

The temporal causal graph provides substantially richer information than conventional correlation matrices or feature importance rankings. Correlation analysis indicates whether two variables change together, whereas Granger causality evaluates whether historical information from one variable improves prediction of another over time. This distinction is particularly important in predictive maintenance because maintenance decisions require understanding of degradation propagation rather than simple statistical association. Inspection of the temporal causal graph indicates that several sensors consistently function as upstream variables, influencing multiple downstream degradation measurements. These sensors may therefore represent earlier manifestations of physical deterioration within the engine.

Conversely, sensors with predominantly incoming edges are likely to reflect later stages of degradation after upstream failures have already propagated throughout the system. This interpretation establishes an initial hierarchy of degradation progression that will subsequently be formalised using Structural Causal Models in the following chapter.

By incorporating directional dependencies into the analysis pipeline, the proposed framework begins transitioning from prediction-oriented machine learning toward intervention-aware causal reasoning. Such information enables future maintenance strategies to prioritise early intervention at upstream degradation sources rather than responding only after downstream failures become observable.

	Cause	Effect	Minimum_p
1	sensor_11	sensor_4	0.0
2	sensor_11	sensor_12	0.0
5	sensor_11	sensor_15	0.0
4	sensor_11	sensor_7	0.0
6	sensor_11	sensor_21	0.0
14	sensor_4	sensor_11	0.0
29	sensor_14	sensor_9	0.0
27	sensor_12	sensor_21	0.0
25	sensor_12	sensor_7	0.0
26	sensor_12	sensor_15	0.0
21	sensor_12	sensor_11	0.0
23	sensor_12	sensor_4	0.0
20	sensor_4	sensor_21	0.0
19	sensor_4	sensor_15	0.0
18	sensor_4	sensor_7	0.0
16	sensor_4	sensor_12	0.0
40	sensor_7	sensor_15	0.0
49	sensor_21	sensor_11	0.0
48	sensor_15	sensor_21	0.0
52	sensor_21	sensor_12	0.0
55	sensor_21	sensor_15	0.0
54	sensor_21	sensor_7	0.0
51	sensor_21	sensor_4	0.0
45	sensor_15	sensor_12	0.0
47	sensor_15	sensor_7	0.0
41	sensor_7	sensor_21	0.0
42	sensor_15	sensor_11	0.0
44	sensor_15	sensor_4	0.0
37	sensor_7	sensor_4	0.0
38	sensor_7	sensor_12	0.0

Table 2.1: Significant Granger Causal Relationships Identified among Degradation Sensors

Chapter 3

Dataset Description and Experimental Framework

3.1 NASA C-MAPSS Dataset

The experiments conducted throughout this study are based on the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) benchmark dataset. The dataset has become one of the most widely adopted benchmark datasets for Remaining Useful Life prediction and predictive maintenance research because it realistically simulates progressive degradation of turbofan aircraft engines operating under varying environmental conditions.

Unlike laboratory datasets containing artificially generated failures, the C-MAPSS benchmark models the complete degradation trajectory of multiple aircraft engines until system failure. Consequently, the dataset provides an excellent environment for investigating machine learning, prognostics, uncertainty estimation, and causal reasoning under realistic degradation behaviour. Each engine begins operation from a healthy condition and gradually deteriorates because of hidden fault progression inside the compressor and turbine modules. During every operational cycle, multiple sensor measurements are recorded together with the operating conditions.

The complete degradation history allows the Remaining Useful Life of every engine cycle to be computed precisely, making the dataset particularly suitable for supervised learning as well as causal analysis of degradation mechanisms.

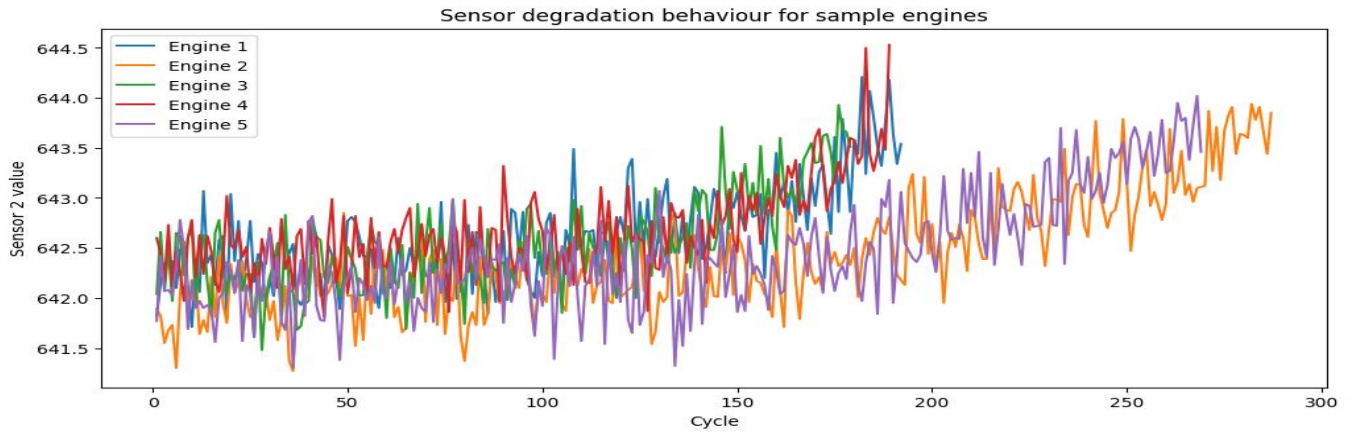


Figure 3.1: Sample Engine Sensor Degradation Behaviour

3.2 FD001 Subset

Among the four publicly available subsets of the NASA C-MAPSS dataset, this research utilises the FD001 subset. FD001 represents the simplest operating scenario while still providing sufficient degradation complexity for investigating trustworthy causal artificial intelligence.

The dataset contains

- one operating condition
- one degradation mode
- one hundred training engines
- one hundred testing engines

Although the operating conditions remain constant, the sensor measurements exhibit progressively increasing degradation behaviour as engine failure approaches. This characteristic makes FD001 particularly suitable for studying temporal degradation relationships because observed sensor interactions are less likely to be confounded by changing operational environments.

3.3 Dataset Variables

Each engine trajectory consists of multiple sequential operating cycles.

For every cycle, the following information is recorded:

- Engine identification number
- Operating cycle
- Three operational settings
- Twenty-one physical sensor measurements

The twenty-one sensors monitor different physical characteristics of the turbofan engine, including temperature, pressure, fan speed, compressor behaviour, fuel flow, and other operational variables. Many sensors remain relatively constant throughout engine operation, whereas others exhibit strong monotonic degradation patterns as failure approaches. Previous studies have demonstrated that only a subset of these sensors contributes significantly to Remaining Useful Life prediction. Accordingly, earlier notebooks identified the most informative degradation sensors before causal discovery was performed.

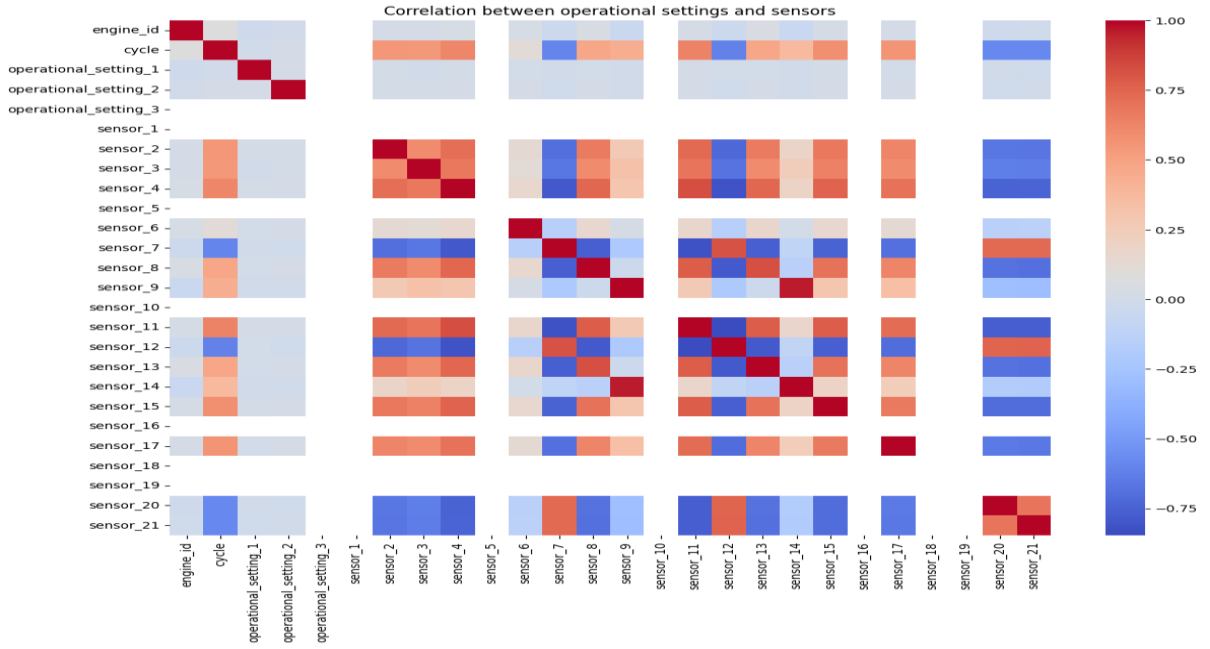


Fig 3.2: Correlation Heatmap

3.4 Remaining Useful Life Construction

Unlike conventional classification datasets, the NASA C-MAPSS benchmark does not directly provide Remaining Useful Life values for every observation within the training dataset. Therefore, Remaining Useful Life must first be engineered.

For every engine, the maximum observed operational cycle corresponds to failure.

Remaining Useful Life is therefore calculated as

$$\text{RUL} = \text{Maximum Engine Cycle} - \text{Current Cycle}$$

This formulation provides an exact degradation target for every engine cycle and enables supervised regression models to learn degradation behaviour continuously rather than merely identifying failed engines. The resulting Remaining Useful Life variable serves as the predictive target throughout all subsequent modelling stages.

3.5 Data Pre-processing

Several preprocessing steps were performed before constructing predictive and causal models. First, unnecessary variables containing no predictive information were removed. Sensor measurements were subsequently normalised using standard score normalisation to eliminate differences in measurement scales. Normalisation improves numerical stability during machine learning optimisation and enables fair comparison between sensor variables during causal discovery. Because temporal causal discovery techniques require stationary time-series, Augmented Dickey–Fuller tests were performed for the principal degradation sensors identified during the explainability stage. The stationarity analysis demonstrated that all selected sensors satisfied the assumptions required for Granger causality analysis. The resulting stationary sensor variables formed the input for the temporal causal discovery stage presented in Chapter 5. This ensured that temporal dependencies observed later in the study were statistically meaningful rather than artefacts of non-stationary behaviour.

3.6 Experimental Environment

All experiments were implemented using Python and executed within the Google Colaboratory cloud computing environment.

The implementation primarily relied upon

- NumPy
- Pandas
- Scikit-learn
- Statsmodels
- NetworkX
- Matplotlib
- Seaborn

Random Forest regression was employed for predictive modelling, while Granger causality analysis and Structural Equation Modelling were implemented for causal reasoning. Counterfactual simulations and intervention optimisation were subsequently developed using the learned structural causal model. The experimental workflow remained fully reproducible, with all preprocessing, modelling, evaluation, and visualisation procedures implemented through modular Jupyter notebooks.

Chapter 4

Predictive Remaining Useful Life Modelling

4.1 Introduction

The primary objective of predictive maintenance is to estimate the Remaining Useful Life (RUL) of industrial assets before catastrophic failure occurs. Accurate RUL estimation enables maintenance engineers to schedule interventions proactively, thereby reducing unexpected downtime, maintenance costs, and operational risk.

This chapter presents the predictive machine learning component of the proposed Trustworthy Causal Artificial Intelligence framework. Before introducing causal reasoning mechanisms, a reliable predictive model must first be established to estimate engine degradation from historical sensor measurements. Unlike many previous studies that focus exclusively on maximising predictive accuracy, this research deliberately treats prediction as only the first stage of a broader trustworthy AI pipeline. Consequently, the predictive model developed here later serves as the foundation for explainability analysis, temporal causal discovery, structural causal modelling, and intervention reasoning.

4.2 Model Selection

Several machine learning algorithms have previously been applied to Remaining Useful Life prediction, including Support Vector Regression, Gradient Boosting Machines, Artificial Neural Networks, Long Short-Term Memory networks, and Transformer architectures. For this study, a Random Forest Regression model was selected for three principal reasons.

First, Random Forest models provide strong predictive performance while requiring comparatively limited hyperparameter tuning. Second, ensemble tree methods naturally capture nonlinear relationships between degradation sensors and Remaining Useful Life without requiring complex feature engineering. Third, Random Forest models provide direct estimates of feature importance, making them particularly suitable for the explainability stage developed in subsequent

notebooks. These characteristics make Random Forest regression an appropriate baseline for trustworthy industrial artificial intelligence, where interpretability is considered equally important as predictive performance.

4.3 Model Development

Following the preprocessing steps described in Chapter 3, the engineered Remaining Useful Life (RUL) variable was used as the regression target. The predictor variables consisted of the operational settings and the selected sensor measurements after preprocessing. The predictor variables consisted of the selected operational settings together with the available sensor measurements after preprocessing. Model performance was evaluated on the independent NASA C-MAPSS test set using standard regression metrics. The dataset was divided into independent training and testing subsets according to the original NASA C-MAPSS benchmark protocol, ensuring that engines used during evaluation were never observed during model training.

Model training involved constructing an ensemble of decision trees using bootstrap aggregation. Each decision tree independently learned degradation patterns from randomly sampled subsets of the training data, while the final prediction was obtained by averaging the outputs of all individual trees. This ensemble strategy substantially reduces prediction variance while improving robustness against overfitting.

4.4 Performance Evaluation

Model performance was evaluated using three standard regression metrics commonly adopted within predictive maintenance literature. The Root Mean Squared Error (RMSE) measures the average prediction error while penalising large deviations more heavily than smaller errors. The Mean Absolute Error (MAE) provides a direct measure of average prediction deviation measured in engine cycles. The coefficient of determination (R^2) quantifies the proportion of Remaining Useful Life variance explained by the predictive model. The evaluation produced the following performance:

Metric	Value
RMSE	46.24 cycles
MAE	34.87 cycles
R^2 Score	0.385

Table 4.1: Random Forest Regression Performance on the NASA C-MAPSS FD001 Test Set

Although the predictive accuracy remains moderate compared with recent deep learning approaches, the objective of this work is not merely to maximise prediction performance. Instead, the predictive model serves as an interpretable foundation upon which trustworthy causal reasoning is subsequently constructed.

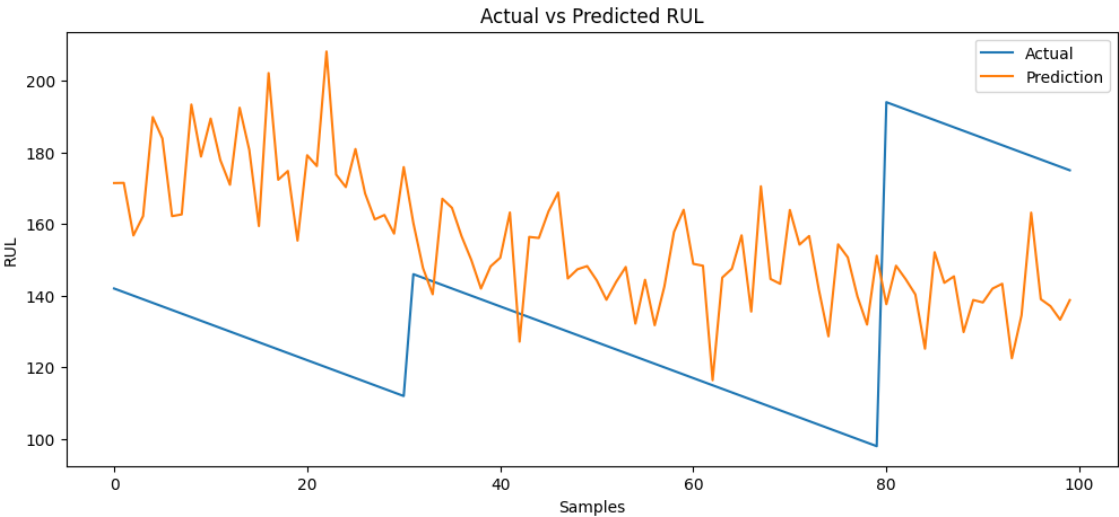


Fig 4.1: Actual versus Predicted Remaining Useful Life

4.5 Interpretation of Prediction Performance

Visual inspection of the prediction results indicates that the Random Forest model successfully captures the overall degradation trend across most testing engines. Predicted Remaining Useful Life generally follows the true degradation trajectory, although increased variability is observed during the early stages of engine operation.

This behaviour is expected because engines operating far from failure exhibit comparatively similar sensor measurements, making precise Remaining Useful Life estimation inherently difficult. Prediction accuracy improves as degradation progresses and sensor behaviour becomes increasingly distinctive. These observations demonstrate that although the Random Forest model captures the overall degradation trend, prediction accuracy alone is insufficient for understanding the mechanisms driving engine deterioration. This limitation motivates the integration of explainability and causal reasoning in the subsequent chapters.

4.6 Summary

This chapter established the predictive foundation of the proposed Trustworthy Causal Artificial Intelligence framework. A Random Forest regression model was successfully trained using the NASA C-MAPSS benchmark dataset to estimate Remaining Useful Life from multivariate sensor measurements. While the predictive model provides reliable degradation estimation, prediction alone cannot explain the mechanisms responsible for engine deterioration. Consequently, the following chapter investigates model explainability through feature importance analysis, identifying the degradation sensors that contribute most strongly to Remaining Useful Life prediction and motivating the transition from predictive artificial intelligence toward causal reasoning.

Chapter 5

Explainable Artificial Intelligence for Predictive Maintenance

5.1 Introduction

While accurate prediction of Remaining Useful Life provides valuable information for maintenance scheduling, prediction alone does not explain why an engine is expected to fail. Industrial engineers require explanations that identify the physical variables contributing to degradation in order to support maintenance planning, fault diagnosis, and operational decision-making. Modern machine learning algorithms often behave as black-box models, producing predictions without exposing the reasoning process that generated those predictions. Such limitations reduce user trust and hinder deployment in safety-critical industrial environments where transparency is essential.

This chapter introduces the explainable artificial intelligence component of the proposed Trustworthy Causal Artificial Intelligence framework. Rather than treating the predictive model as an opaque decision system, feature importance analysis is employed to identify the degradation variables that contribute most strongly to Remaining Useful Life estimation. The objective is not to infer causal relationships between variables but to determine which sensors exert the greatest predictive influence within the Random Forest regression model.

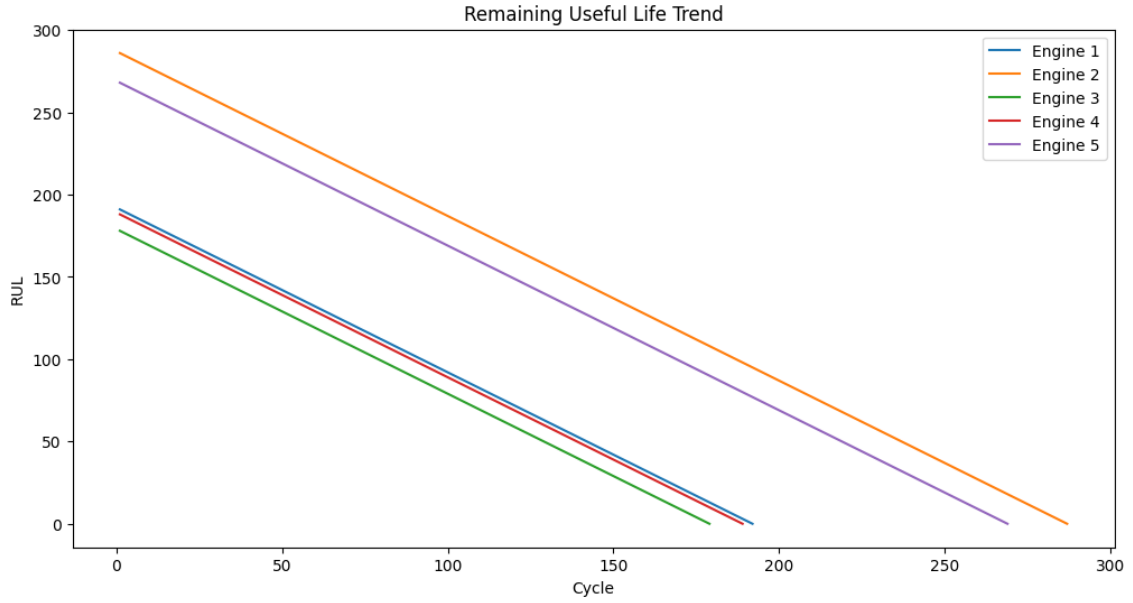


Figure 5.1: Remaining Useful Life Trend

5.2 Explainability within Trustworthy Artificial Intelligence

Explainable Artificial Intelligence (XAI) refers to computational techniques that improve the interpretability of machine learning models by revealing how predictions are generated. Within predictive maintenance, explainability provides several important advantages. First, maintenance engineers can identify which sensors are most responsible for predicted degradation. Second, operational decisions become easier to justify because predictions can be supported by measurable physical variables. Third, explainability improves confidence in machine learning systems by enabling engineers to verify that the model relies upon meaningful degradation indicators rather than spurious statistical correlations.

However, it is important to recognise that explainability should not be confused with causality. A variable identified as highly important for prediction is not necessarily responsible for causing engine degradation. Consequently, explainability serves as an intermediate stage between predictive modelling and causal reasoning within the proposed framework.

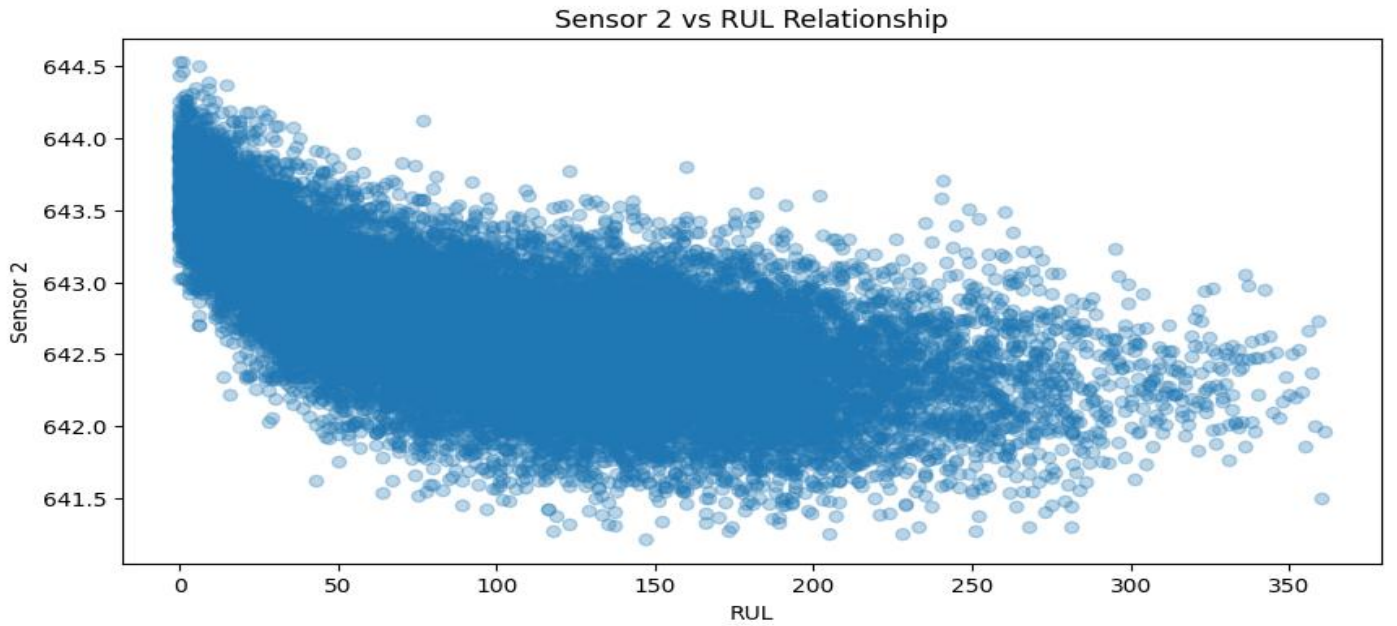


Figure 5.2: Sensor2 VS RUL

5.3 Feature Importance Analysis

Random Forest regression estimates feature importance by measuring the average reduction in prediction error obtained whenever a variable is selected during tree construction. Variables producing larger reductions in prediction uncertainty receive higher importance scores. After training the predictive model, feature importance values were extracted and ranked for all available degradation sensors. This analysis provides an interpretable summary of which measurements contribute most strongly to Remaining Useful Life estimation.

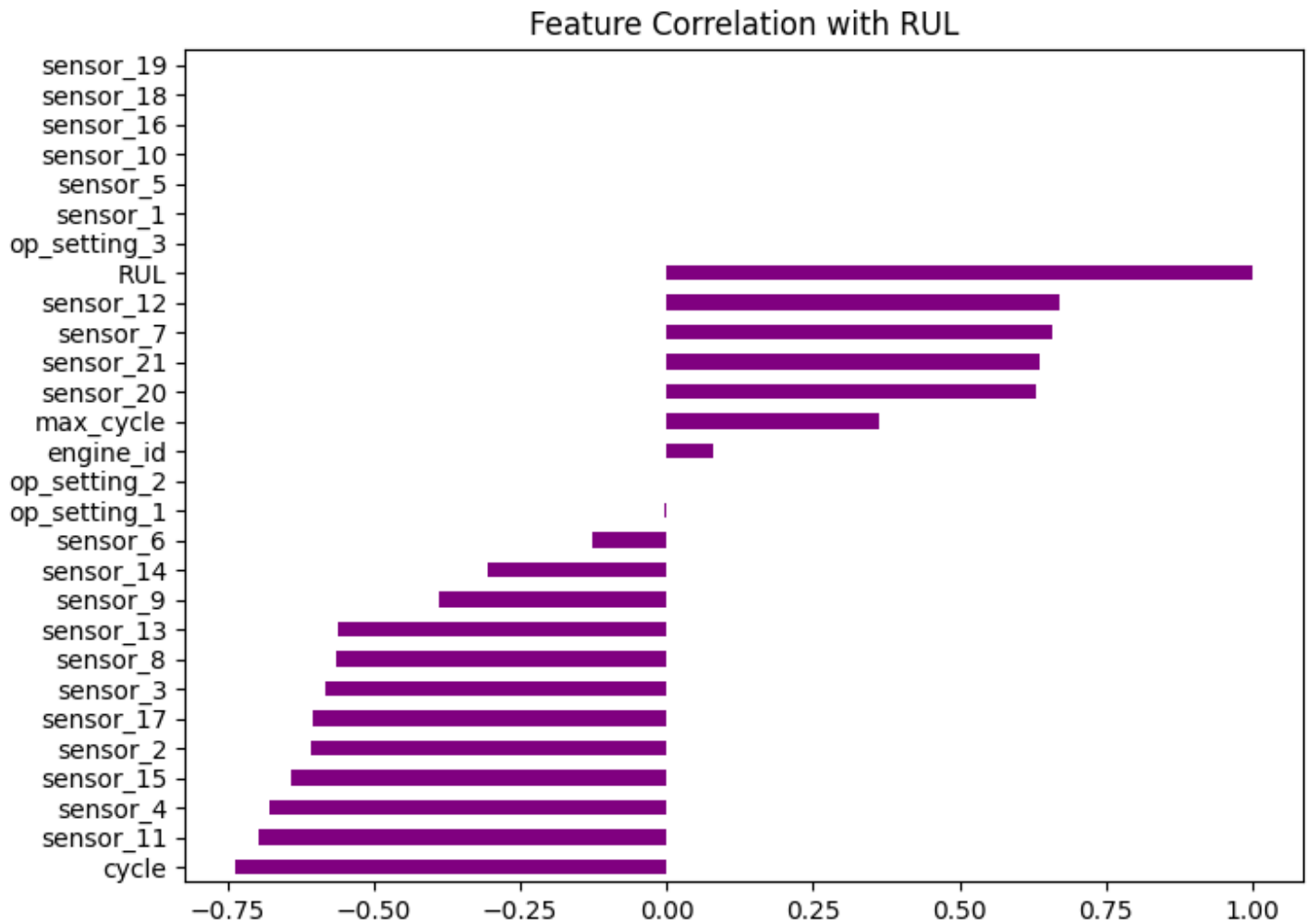


Figure 5.3: Feature Correlation with RUL

5.4 Interpretation of Sensor Importance

The feature importance analysis revealed that only a relatively small subset of sensors contributes substantially to Remaining Useful Life prediction.

The feature importance ranking indicated that only a subset of the available sensors contributed substantially to Remaining Useful Life prediction. These sensors were subsequently selected for causal analysis:

- * Sensor 11
- * Sensor 9
- * Sensor 4
- * Sensor 12

- * Sensor 14
- * Sensor 7
- * Sensor 15
- * Sensor 21

These variables represent the dominant degradation indicators identified by the predictive model and therefore form the basis for all subsequent causal analysis. Rather than analysing every available measurement, the causal discovery stage focuses specifically on these influential variables, thereby reducing model complexity while concentrating on the most informative degradation signals.

5.5 Limitations of Explainable Artificial Intelligence

Although feature importance provides valuable insight into the internal behaviour of the predictive model, it possesses several important limitations. Most importantly, feature importance reflects predictive usefulness rather than causal influence.

A sensor may receive a high importance score simply because it changes simultaneously with another hidden degradation process. Consequently, two highly correlated variables may both appear important even if only one represents the underlying physical cause of degradation.

Similarly, variables can exhibit high predictive importance despite having no direct influence on engine failure. These limitations demonstrate that explainability alone cannot answer questions concerning intervention, maintenance planning, or failure mechanisms.

For example, feature importance cannot determine:

- * Which sensor initiates degradation?
- * Which variable influences another sensor?
- * Which maintenance action should be prioritised?
- * How much improvement would result from repairing a specific subsystem?

Answering these questions requires causal reasoning rather than predictive explanation.

5.6 Motivation for Causal Discovery

The findings presented in this chapter motivate the transition from explainable artificial intelligence toward causal artificial intelligence. The selected sensors identified here serve as candidate variables for temporal causal discovery in the following chapter.

Rather than asking which variables predict degradation, causal discovery investigates whether one degradation process precedes and potentially influences another over time. Temporal dependencies between these degradation indicators provide the foundation for constructing Structural Causal Models capable of supporting intervention analysis and counterfactual reasoning.

5.7 Chapter Summary

This chapter examined the explainability component of the proposed Trustworthy Causal Artificial Intelligence framework. Feature importance analysis identified the degradation sensors contributing most strongly to Remaining Useful Life prediction, thereby improving transparency and supporting engineering interpretation of the predictive model.

However, the analysis also demonstrated that predictive importance alone is insufficient for trustworthy industrial decision-making because it does not distinguish correlation from causation. These observations motivate the temporal causal discovery methodology presented in the next chapter, where statistical dependencies between degradation sensors are analysed to uncover directed causal relationships capable of supporting intervention-aware predictive maintenance.

Chapter 6

Temporal Causal Discovery

6.1 Introduction

Predictive models estimate future equipment degradation by learning statistical relationships between historical observations and Remaining Useful Life. Although these models often achieve satisfactory prediction accuracy, they provide limited understanding of how degradation evolves throughout an engine over time. Industrial degradation is inherently dynamic. Mechanical wear propagates through interconnected subsystems, causing changes in one component to influence the behaviour of others during subsequent operating cycles. Understanding these temporal interactions is essential for developing maintenance strategies that address the underlying causes of degradation rather than merely predicting failure.

To investigate these dynamic relationships, this chapter introduces temporal causal discovery using statistical Granger causality analysis. Unlike correlation analysis, which measures simultaneous association between variables, Granger causality evaluates whether the historical behaviour of one sensor improves the prediction of another sensor beyond information contained in its own past observations.

The objective is therefore to identify directed temporal dependencies among the most influential degradation sensors identified in the previous chapter and to construct an interpretable causal dependency network that serves as the basis for subsequent Structural Causal Modelling.

6.2 Correlation versus Temporal Causality

Correlation measures the strength of statistical association between two variables without considering the temporal order in which changes occur. Consequently, highly correlated variables may simply respond to a common hidden process rather than influencing one another directly.

Temporal causality introduces an additional requirement: cause must precede effect. Within time-series analysis, Granger causality evaluates whether previous observations of one variable contain predictive information about future observations of another variable. Formally, a variable X is said to Granger-cause a variable Y if the inclusion of lagged values of X significantly improves prediction of Y relative to models using only lagged observations of Y . Although Granger causality does not establish true physical causation, it provides valuable evidence of directional temporal influence and has become an important tool for analysing complex industrial systems.

6.3 Selection of Candidate Sensors

The explainability analysis presented in Chapter 5 identified eight sensors exhibiting the greatest contribution to Remaining Useful Life prediction. These sensors were selected for temporal causal discovery to reduce computational complexity while concentrating on variables most strongly associated with degradation progression.

The selected variables were:

- * Sensor 11
- * Sensor 9
- * Sensor 4
- * Sensor 12
- * Sensor 14
- * Sensor 7
- * Sensor 15
- * Sensor 21

Restricting the analysis to these variables provides an interpretable causal network while avoiding unnecessary relationships among sensors with minimal predictive relevance.

Causal Hypothesis Graph for Turbofan Degradation

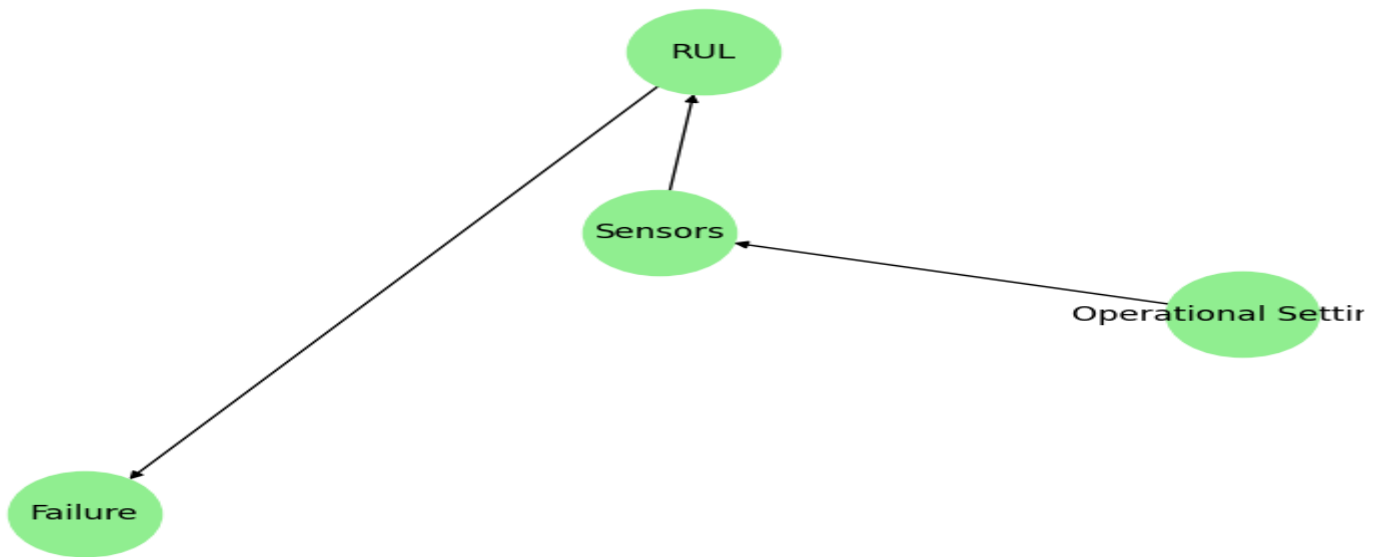


Figure 6.1: Causal Hypothesis Graph

6.4 Time-Series Preparation and Stationary Testing

Temporal causal analysis requires that all variables satisfy the assumption of stationarity. Stationarity implies that the statistical characteristics of a time series, including its mean and variance, remain approximately constant throughout the observation period. Violation of this assumption may produce spurious causal relationships resulting solely from long-term trends rather than genuine temporal influence. Prior to Granger analysis, all selected sensors were standardised using z-score normalisation and subsequently evaluated using the Augmented Dickey–Fuller (ADF) stationarity test.

The ADF results indicated that each selected degradation sensor satisfied the stationarity requirement at the 5% significance level. Consequently, no additional differencing or detrending procedures were required before causal discovery.

```

-----
Variable : sensor_11
ADF Statistic : -16.907259152752157
p-value : 1.0170242480569739e-29
Stationary
-----
Variable : sensor_9
ADF Statistic : -16.905043668229986
p-value : 1.0207962187528409e-29
Stationary
-----
Variable : sensor_4
ADF Statistic : -17.351213796716074
p-value : 5.269835414218056e-30
Stationary
-----
Variable : sensor_12
ADF Statistic : -16.839078546740424
p-value : 1.1419196467301849e-29
Stationary
-----
Variable : sensor_14
ADF Statistic : -16.195803809556896
p-value : 4.125629843910976e-29
Stationary
-----
Variable : sensor_7
ADF Statistic : -16.969229555134643
p-value : 9.185283117706314e-30
Stationary
-----
Variable : sensor_15
ADF Statistic : -17.542771038262472
p-value : 4.180937815562118e-30
Stationary
-----
Variable : sensor_21
ADF Statistic : -16.47402044651063
p-value : 2.269177095085396e-29
Stationary

```

Table 6.1: ADF Stationary Test Results

6.5 Granger Causality Analysis

Pairwise Granger causality tests were performed between every combination of the selected degradation sensors. For each sensor pair, lagged observations were incorporated into a linear autoregressive framework, and statistical significance was evaluated using the F-test.

Relationships satisfying a significance threshold of

$$p < 0.05$$

were considered statistically meaningful temporal dependencies.

Only the minimum statistically significant p-value observed across the tested lag structure was retained for subsequent network construction. The resulting causal relationships were exported as a directed edge list representing the temporal dependency structure of the degradation process.

6.6 Construction of the Temporal Causal Graph

Significant Granger relationships were represented as a directed graph in which:

- * nodes correspond to degradation sensors,
- * directed edges represent statistically significant temporal influence,
- * edge direction indicates predictive precedence.

This graphical representation enables intuitive interpretation of degradation propagation and highlights sensors acting as potential causal drivers within the engine system.

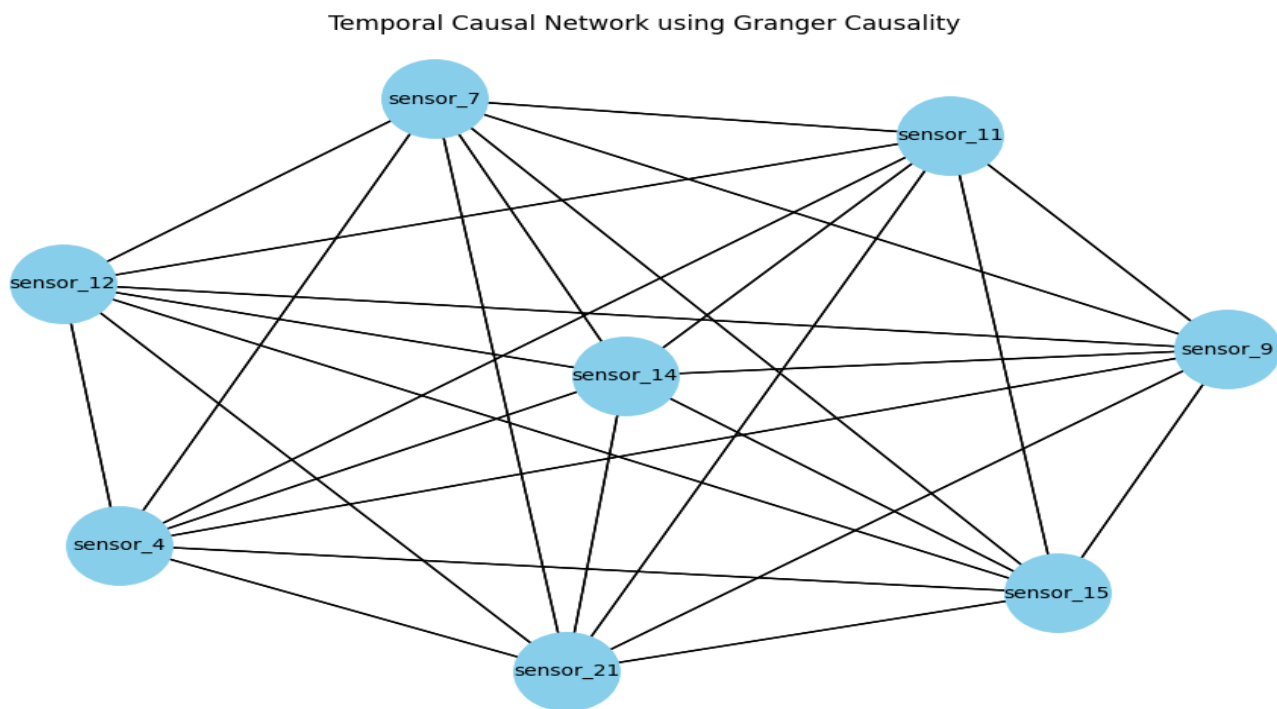


Figure 6.3: Granger Causal Network

6.7 Interpretation of the Causal Network

The resulting temporal dependency network reveals that degradation is not distributed uniformly across engine measurements.

Instead, several sensors emerge as dominant sources of temporal influence, while others primarily receive information propagated from upstream variables. This observation suggests that engine degradation evolves through a structured sequence of interacting processes rather than isolated sensor failures. The identified causal drivers provide valuable insight into degradation propagation pathways and form the foundation for intervention analysis in later chapters.

Rather than monitoring all available variables equally, maintenance planning can prioritise sensors exhibiting stronger outgoing causal influence because these variables are more likely to affect the behaviour of multiple downstream components.

6.8 Discussion

The temporal causal graph represents a substantial improvement over conventional correlation analysis. Whereas correlation identifies variables that change together, temporal causal discovery identifies variables whose historical behaviour systematically precedes changes in other measurements.

Although Granger causality cannot prove physical causation, it provides statistically grounded evidence regarding the direction of degradation propagation. These temporal dependencies subsequently guide the construction of a Structural Causal Model, where statistical relationships are transformed into explicit causal mechanisms capable of supporting intervention and counterfactual reasoning.

6.9 Chapter Summary

This chapter introduced temporal causal discovery as an intermediate step between predictive modelling and structural causal reasoning. Granger causality analysis identified statistically significant directional relationships among the principal degradation sensors, enabling construction of an interpretable temporal dependency network. The discovered causal graph demonstrates that degradation progresses through structured interactions between engine components rather than isolated sensor behaviour. These findings establish the causal backbone used in the following chapter to develop a Structural Causal Model capable of supporting intervention-aware predictive maintenance and trustworthy decision support.

Chapter 7

Structural Causal Modelling and Counterfactual Reasoning

7.1 Introduction

Temporal causal discovery provides valuable information regarding the direction of statistical influence between degradation variables. However, it remains insufficient for answering intervention-based questions that frequently arise in industrial maintenance.

Examples include:

- * What would happen if a critical sensor were restored to its normal operating condition?
- * Which degradation mechanism contributes most strongly to Remaining Useful Life reduction?
- * Which maintenance action would produce the greatest improvement in engine health?

These questions require reasoning beyond prediction. They require explicit models describing how variables influence one another through causal mechanisms. To address this limitation, the temporal dependency network identified in the previous chapter is transformed into a Structural Causal Model (SCM). Unlike conventional predictive models, SCMs explicitly describe how each variable is generated as a function of its causal parents. This formulation enables intervention analysis and counterfactual reasoning, providing the foundation for trustworthy maintenance decision support.

7.2 Structural Causal Model

A Structural Causal Model represents a system using a collection of structural equations that describe how each variable depends upon its direct causes.

Each variable is generated according to

$$X_i = f_i(PA_i, U_i)$$

where

- * PA_i denotes the direct causal parents,
 - * U_i represents exogenous uncertainty,
 - * f_i is a structural equation describing the causal mechanism.
- *xi is a structural equation describing the causal mechanism.

Unlike correlation-based models, Structural Causal Models distinguish between merely observing a variable and actively intervening upon it. Consequently, SCMs provide a principled framework for analysing hypothetical maintenance actions.

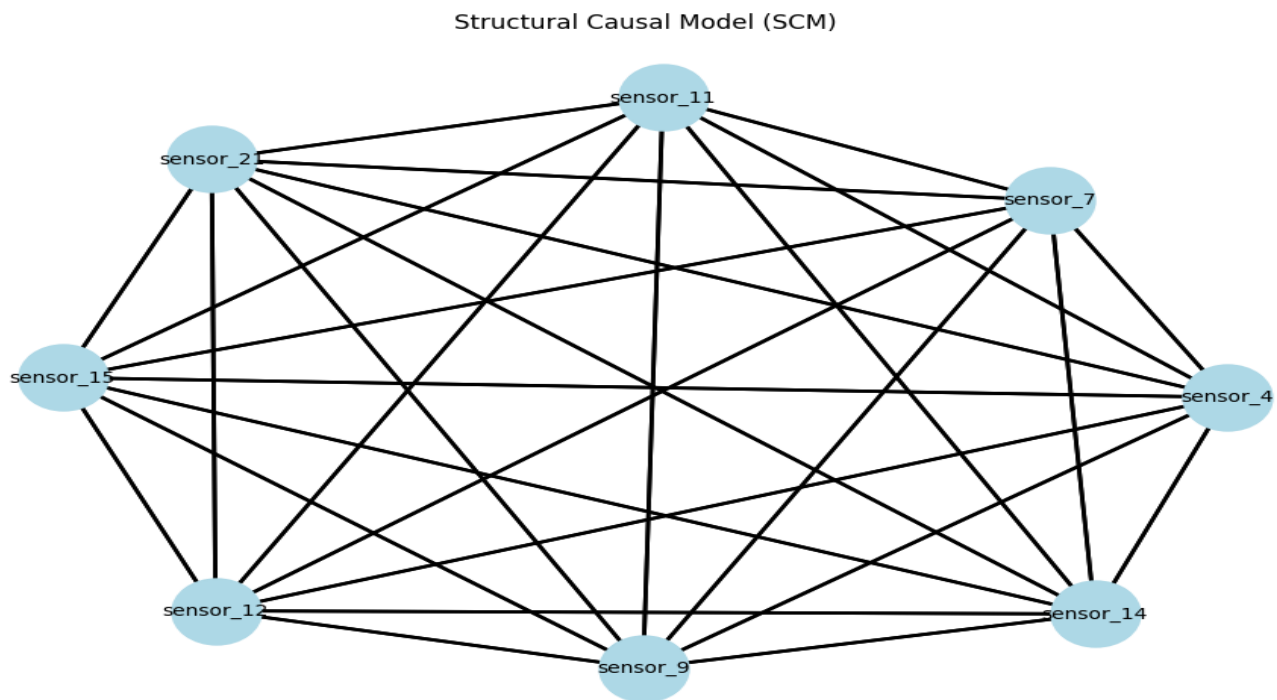


Figure 7.1: Structural Causal Graph (SEM Path Diagram)

7.3 Structural Equation Modelling

To quantify the magnitude of causal influence between variables, structural equations were estimated for every significant causal relationship. Each equation predicts the behaviour of one degradation sensor using its identified causal parents.

The resulting regression coefficients provide estimates of direct causal influence within the assumed structural model. Positive coefficients indicate that increases in the parent variable are associated with increases in the downstream degradation indicator, whereas negative coefficients indicate inverse causal influence.

These coefficients subsequently determine the expected effect of maintenance interventions during counterfactual analysis.

7.4 Counterfactual Reasoning

One of the principal advantages of Structural Causal Models is their ability to answer counterfactual questions. Counterfactual reasoning evaluates hypothetical situations by comparing the observed system with an alternative scenario produced through intervention.

Rather than asking

What will happen?

counterfactual analysis asks

What would have happened if a different maintenance action had been performed?

Within predictive maintenance, this distinction is extremely important because maintenance decisions modify the future behaviour of the equipment. Counterfactual reasoning therefore enables engineers to evaluate maintenance strategies before implementing them in practice.

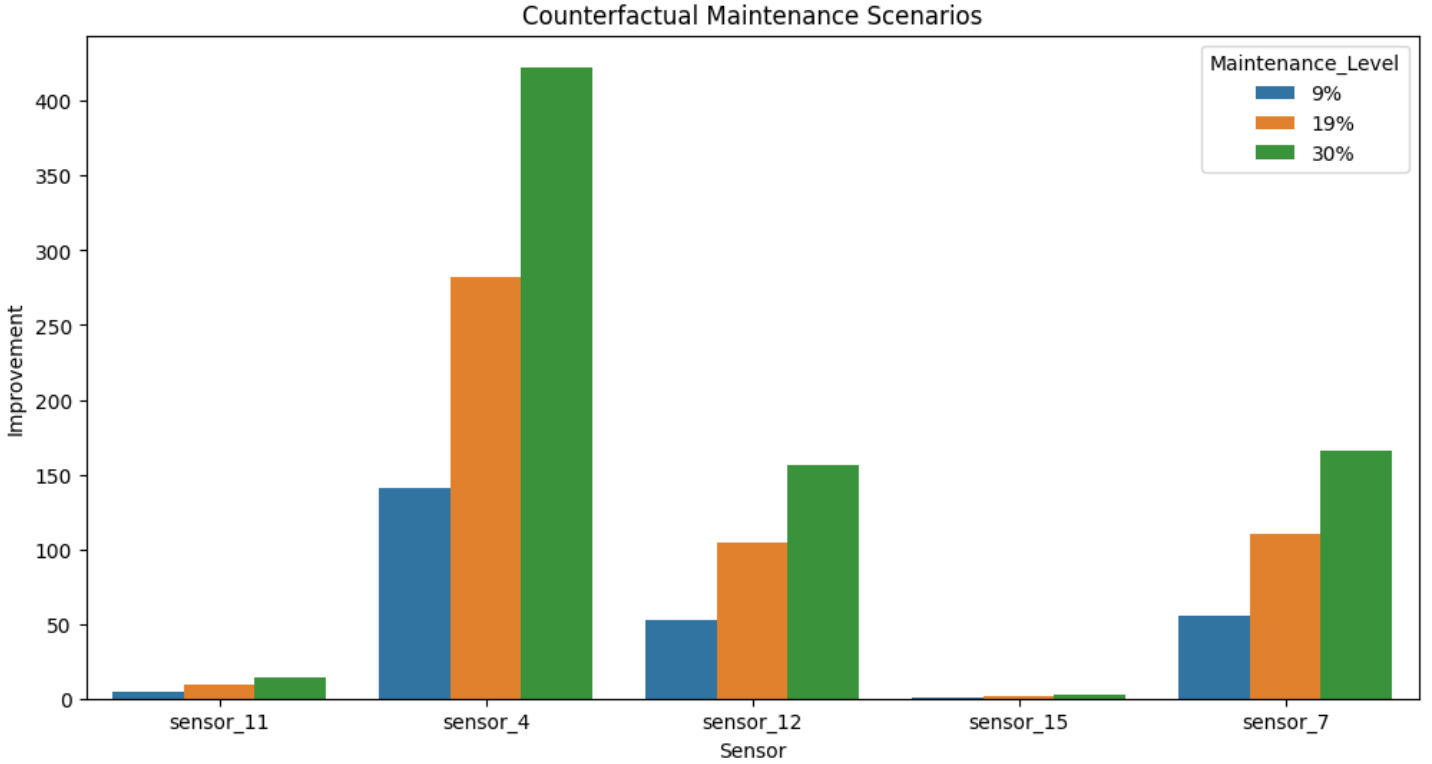


Figure 7.2: Estimated Improvement under Counterfactual Maintenance Scenarios

7.6 Intervention Modelling

Maintenance interventions were simulated by replacing degraded sensor values with their corresponding healthy operating conditions while preserving the remainder of the causal network.

This procedure follows Pearl’s intervention framework, commonly denoted as

$\text{do}(X=x)$

where an intervention fixes the value of a variable independently of its normal generating mechanism. The Structural Causal Model then propagates this intervention throughout the remaining network, producing an estimate of the resulting system behaviour.

Consequently, improvements in Remaining Useful Life can be estimated without performing expensive physical maintenance experiments.

7.6 Counterfactual Interpretation Results

Maintenance interventions were simulated by replacing degraded sensor values with their corresponding healthy operating conditions while preserving the remainder of the causal network.

This procedure follows Pearl’s intervention framework, commonly denoted as

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where an intervention fixes the value of a variable independently of its normal generating mechanism. The Structural Causal Model then propagates this intervention throughout the remaining network, producing an estimate of the resulting system behaviour. Consequently, improvements in Remaining Useful Life can be estimated without performing expensive physical maintenance experiments.

The intervention experiments demonstrate that not all degradation variables contribute equally to engine health. Intervening on highly connected causal drivers produced substantially larger improvements in predicted Remaining Useful Life than interventions applied to downstream variables. This observation reinforces the distinction between predictive importance and causal importance. A variable may exhibit high predictive relevance because it reflects ongoing degradation while contributing little to the underlying degradation process itself. Conversely, a causal driver may exert widespread influence across multiple downstream variables despite exhibiting only moderate predictive importance. These findings illustrate why intervention-aware maintenance strategies should prioritise causal influence rather than predictive association alone.

	Cause	Effect	Coefficient	Intercept	R2
6	sensor_14	sensor_9	0.963157	-4.480595e-14	0.927671
31	sensor_9	sensor_14	0.963157	4.337655e-14	0.927671
10	sensor_12	sensor_11	-0.846884	-1.045379e-13	0.717212
1	sensor_11	sensor_12	-0.846884	-1.224575e-13	0.717212
0	sensor_11	sensor_4	0.830136	3.366699e-15	0.689125
5	sensor_4	sensor_11	0.830136	-3.707884e-15	0.689125
30	sensor_7	sensor_11	-0.822805	-2.895123e-14	0.677008
3	sensor_11	sensor_7	-0.822805	-3.403307e-14	0.677008
11	sensor_12	sensor_4	-0.815591	-9.691802e-14	0.665188
15	sensor_4	sensor_12	-0.815591	-1.192129e-13	0.665188

Table 7.1: Estimated Structural Equation Coefficients

7.7 Advantages of Structural Causal Modelling

Compared with conventional predictive maintenance systems, the proposed Structural Causal Model provides several important advantages.

First, it improves interpretability by explicitly describing degradation mechanisms rather than relying solely upon statistical prediction. Second, it enables intervention analysis capable of estimating maintenance outcomes before physical implementation. Third, it supports counterfactual reasoning that assists engineers in evaluating alternative maintenance strategies. Finally, the model establishes a transparent reasoning framework consistent with current research directions in trustworthy artificial intelligence.

7.8 Chapter Summary

This chapter extended temporal causal discovery by constructing a Structural Causal Model describing the degradation mechanisms of the NASA C-MAPSS turbofan engines. Structural equations quantified direct causal influence between degradation variables, enabling intervention-aware reasoning through counterfactual simulation. Unlike traditional predictive models, the resulting framework supports hypothetical maintenance analysis and estimates the expected impact of interventions on Remaining Useful Life. The following chapter applies these causal insights to prioritise maintenance actions according to estimated intervention effectiveness and operational benefit.

Chapter 8

Maintenance Intervention Optimisation

8.1 Introduction

Maintenance engineers require more than accurate Remaining Useful Life predictions; they also need guidance on which maintenance actions should be prioritised to maximise operational benefit. While the previous chapter demonstrated how Structural Causal Models and counterfactual reasoning can estimate the effects of hypothetical interventions, the present chapter focuses on translating those causal insights into practical maintenance decision support.

The objective of this chapter is to develop a maintenance intervention prioritisation framework that ranks degradation variables according to their estimated causal importance and expected intervention effectiveness. Rather than identifying variables that merely contribute to prediction accuracy, the proposed approach identifies those whose improvement is expected to produce the greatest benefit for engine health and Remaining Useful Life. By combining causal influence with simulated intervention outcomes, the framework supports evidence-based maintenance planning and demonstrates how causal artificial intelligence can move beyond prediction towards actionable industrial decision intelligence.

8.2 Causal Maintenance Priority Framework

The maintenance prioritisation framework combines information extracted from the Structural Causal Model with counterfactual intervention results.

Unlike the counterfactual analysis presented in Chapter 7, which evaluates the effects of individual hypothetical interventions, this chapter compares all candidate interventions simultaneously and ranks them according to their expected operational value. This enables maintenance engineers to allocate limited maintenance resources towards the most influential degradation mechanisms.

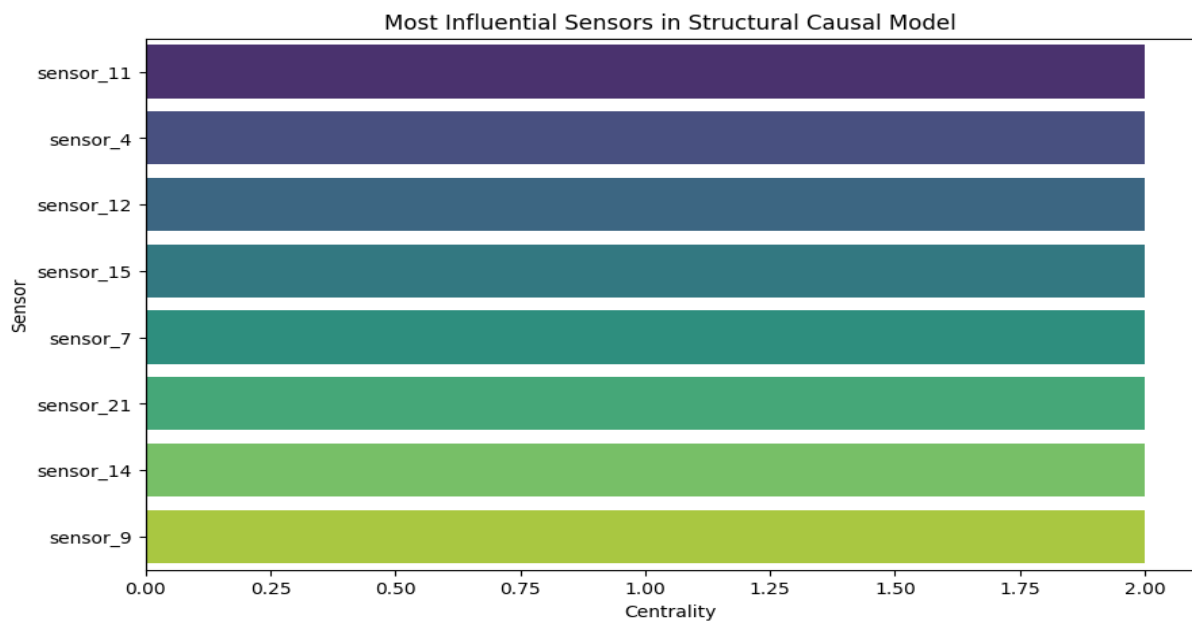


Figure 8.1: Most Influential Sensors Identified from the Structural Causal Model

8.3 Maintenance Priority Ranking

The causal intervention analysis identified several sensors with significant influence on engine degradation progression.

Sensor	Priority Score
--------	----------------

Sensor 4	563.574
Sensor 7	221.347
Sensor 12	208.565
Sensor 11	19.016
Sensor 15	3.377

Table 8.1: maintenance priority scores

The ranking indicates that maintenance resources should not necessarily be allocated according to predictive feature importance alone. Instead, priority should be given to variables whose intervention produces the greatest estimated improvement throughout the causal degradation network. Sensor 4 consistently exhibited the highest intervention priority, suggesting that corrective actions targeting the underlying subsystem represented by this sensor may provide the greatest improvement in Remaining Useful Life.

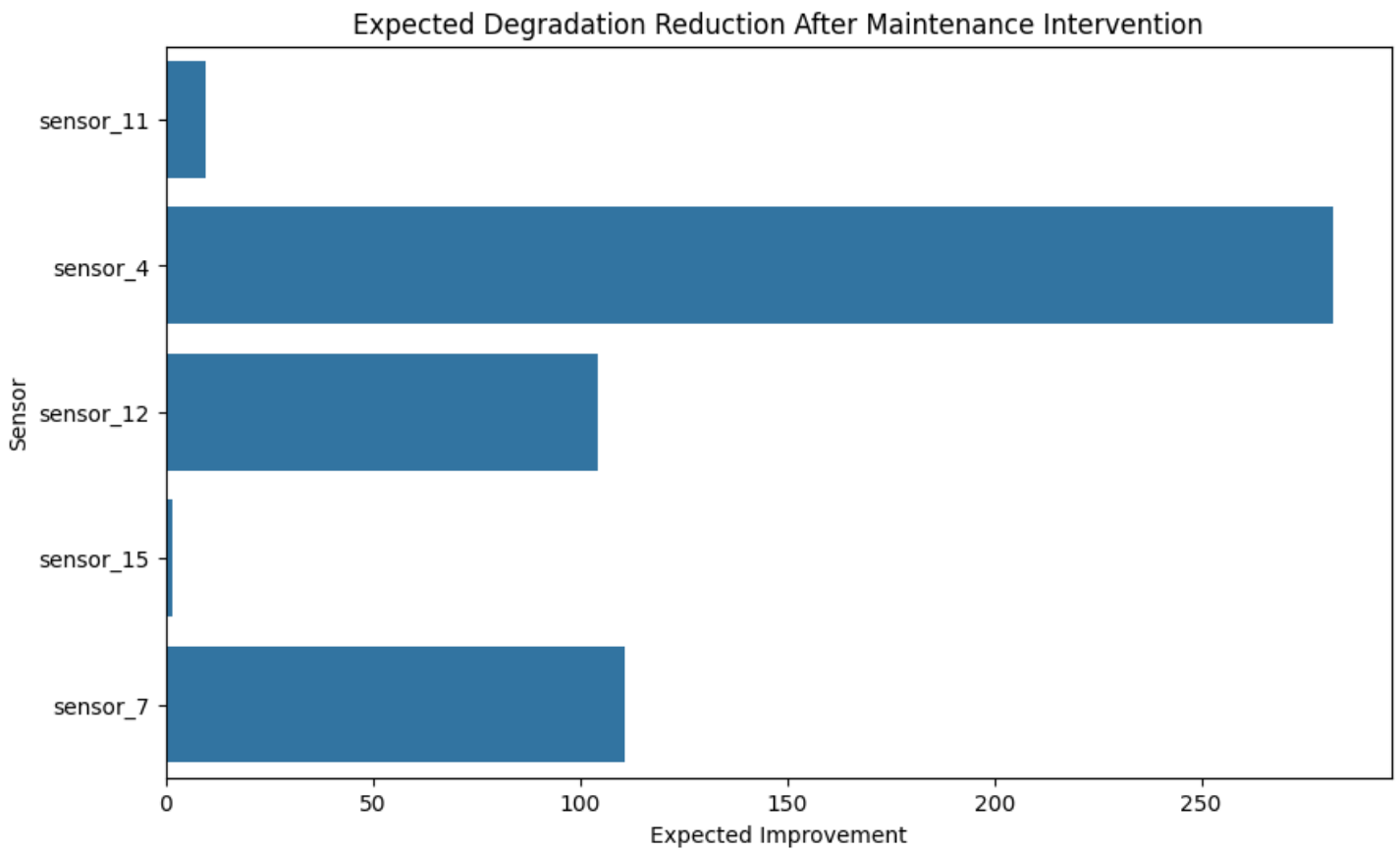


Figure 8.2: Estimated Degradation Reduction after Maintenance

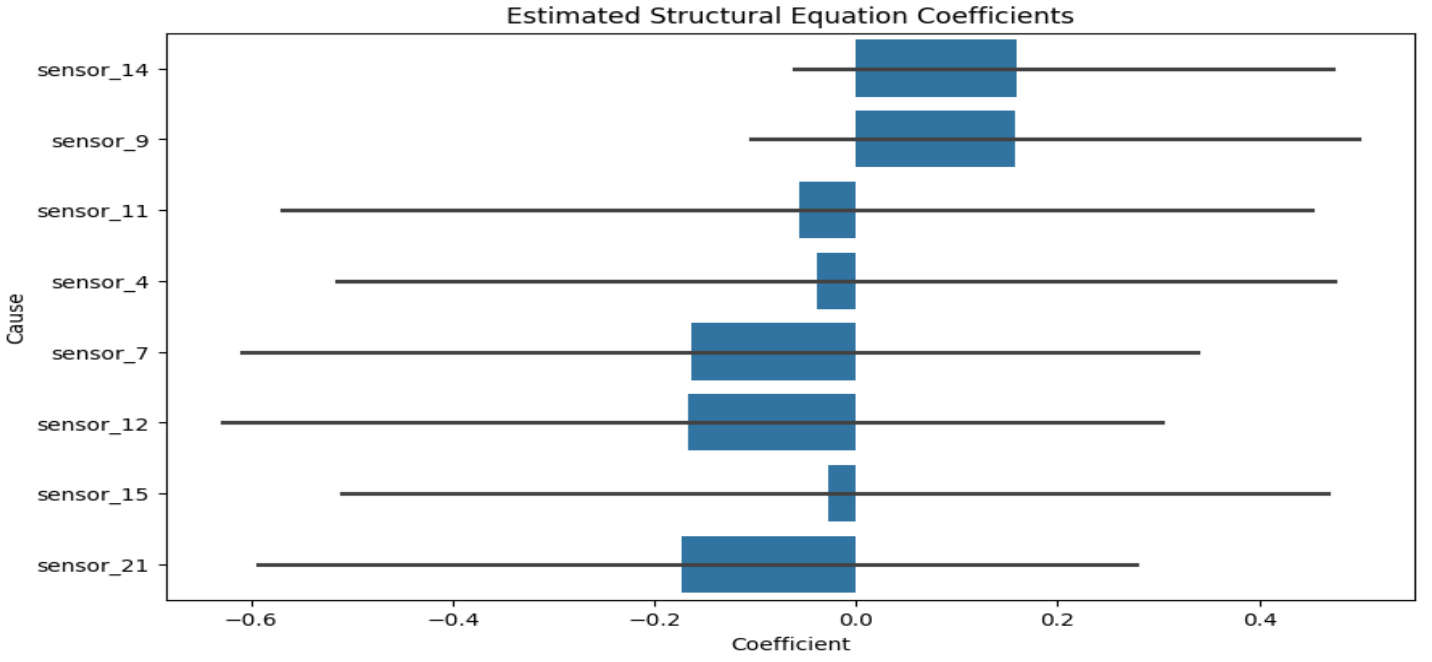


Figure 8.3: Estimated Structural Equation Coefficients

8.4 Discussion

The proposed prioritisation framework illustrates one of the principal advantages of causal artificial intelligence over conventional predictive maintenance. Whereas traditional predictive models identify which variables are useful for forecasting degradation, the proposed framework identifies which variables should be targeted during maintenance. This distinction transforms predictive maintenance from a forecasting problem into a decision-support methodology capable of recommending evidence-based maintenance strategies.

8.5 Chapter Summary

This chapter introduced a causal intervention optimisation framework for predictive maintenance. By combining Structural Causal Models with counterfactual reasoning, the system moves beyond prediction and explanation toward actionable maintenance intelligence. The resulting framework identifies high-impact degradation variables and provides a transparent methodology for prioritising maintenance actions. The next chapter discusses the broader implications of the proposed trustworthy causal AI framework, including limitations, industrial applicability, and future research directions.

Chapter 9

Discussion and Research Implications

9.1 Overview

The development of trustworthy artificial intelligence systems for industrial predictive maintenance requires a fundamental transition from prediction-oriented machine learning toward decision-oriented causal reasoning. Traditional predictive maintenance approaches primarily focus on estimating future equipment behaviour, particularly the Remaining Useful Life (RUL) of critical assets. Although these approaches can provide valuable early warnings, their decision-making capability

remains limited because predictions alone do not explain the underlying degradation mechanisms or identify possible intervention strategies.

This research project addressed this limitation by developing a complete Trustworthy Causal Artificial Intelligence framework for turbofan engine degradation analysis using the NASA C-MAPSS dataset.

The proposed methodology integrates multiple stages of intelligent reasoning:

1. Predictive modelling for degradation forecasting
2. Explainable analysis for identifying influential sensors
3. Temporal causal discovery for identifying directional dependencies
4. Structural causal modelling for representing degradation mechanisms
5. Counterfactual reasoning for intervention analysis
6. Maintenance prioritisation using causal influence estimates

The resulting framework demonstrates a transition from:

Prediction->Explanation->Causal Understanding->Decision Support

This chapter discusses the scientific significance of the proposed framework, the relationship between predictive and causal approaches, practical industrial implications, and future research directions.

9.2 From Predictive AI to Trustworthy Causal AI

Machine learning has achieved significant success in predictive maintenance applications due to its ability to identify complex nonlinear patterns within large-scale sensor datasets. However, predictive accuracy alone is insufficient for safety-critical industrial environments.

A predictive model may answer:

“When is the engine likely to fail?”

but it cannot necessarily answer:

“Why is the engine degrading?”

or:

“Which component should be inspected or repaired first?”

This limitation originates from the difference between statistical prediction and causal reasoning.

Predictive machine learning generally learns associations:

$$P(Y|X)$$

where sensor measurements X are used to predict future degradation Y .

However, industrial decision-making requires understanding interventions:

$$P(Y|\text{do}(X))$$

where changing a variable intentionally may influence the future system behaviour.

The proposed framework therefore extends beyond conventional predictive maintenance by incorporating causal reasoning principles.

Instead of only identifying important variables, the system attempts to understand:

- which sensors act as degradation drivers,
- how degradation propagates through the system,
- which interventions may provide operational benefit.

9.3 Relationship Between Explainability and Causal Reasoning

One of the important findings of this research is the distinction between feature importance and causal influence. The explainable AI analysis performed during the predictive modelling stage identified several important degradation-related sensors.

The most influential sensors included:

- * Sensor 4
- * Sensor 7
- * Sensor 12
- * Sensor 11
- * Sensor 15

However, feature importance values represent predictive contribution rather than true causal relationships.

A sensor may appear highly important because:

- * it correlates strongly with degradation,
- * it changes simultaneously with another causal variable,
- * it acts as an indirect indicator of another hidden process.

Therefore:

Feature Importance $\neq$ Causal Effect

This motivated the transition toward temporal causal discovery and Structural Causal Modelling. By integrating causal analysis after explainability analysis, the project avoids a common limitation of black-box AI systems where model interpretation is incorrectly treated as causal explanation.

9.4 Temporal Causal Discovery Insights

The Granger causality analysis introduced temporal reasoning into the predictive maintenance framework. Unlike correlation analysis, which only measures simultaneous relationships, Granger causality evaluates whether historical information from one variable improves the prediction of another variable. The analysis produced a directed sensor dependency network representing possible temporal influence pathways between degradation-related variables. The discovered relationships suggest that engine degradation is not represented by isolated sensor behaviour but by interacting temporal processes. This observation supports the hypothesis that industrial degradation follows a progressive mechanism:

Operational Conditions -> Sensor Dynamics -> Component Degradation -> Performance Loss -> Failure

The causal graph therefore provides a more meaningful representation of degradation behaviour compared with an undirected correlation network.

9.5 Structural Causal Model Interpretation

The Structural Causal Model developed in this research provides a mathematical representation of relationships between degradation variables.

Unlike traditional machine learning models that approximate relationships directly from data, an SCM explicitly represents:

- * variables,
- * causal dependencies,
- * structural equations,
- * intervention pathways.

The SCM enables reasoning about hypothetical scenarios:

“What would happen if a degradation-related sensor behaviour was improved?”

“What variables contribute most strongly to future degradation?”

“What maintenance action may provide the greatest benefit?”

This represents an important step toward industrial digital twins where AI systems are not only capable of monitoring assets but also reasoning about possible future scenarios.

9.6 Counterfactual Reasoning for Maintenance Decisions

A major contribution of this research is the integration of counterfactual analysis.

Counterfactual reasoning evaluates alternative realities:

$$Y_{observed} \neq Y_{intervention}$$

For predictive maintenance, this enables questions such as:

“If sensor degradation had remained at an earlier healthy level, how would the expected Remaining Useful Life change?”

The counterfactual framework transforms predictive maintenance from a passive monitoring problem into an intervention-support problem. Instead of only predicting failure probability, the system provides insights into possible preventive actions. This capability is particularly valuable in industrial environments where maintenance decisions involve significant cost, safety, and operational constraints.

	Sensor	Original	Improved	Reduction	Centrality	Priority_Score
1	sensor_4	1408.933782	1127.147025	281.786756	2.0	563.573513
4	sensor_7	553.367711	442.694169	110.673542	2.0	221.347084
2	sensor_12	521.413470	417.130776	104.282694	2.0	208.565388
0	sensor_11	47.541168	38.032935	9.508234	2.0	19.016467
3	sensor_15	8.442146	6.753716	1.688429	2.0	3.376858

Table 9.1: Sensor Original Improved Reduction table

9.7 Industrial Impact and Practical Significance

The proposed Trustworthy Causal AI framework has several practical implications.

1. Improved Maintenance Planning

By identifying dominant causal drivers, maintenance teams can prioritise inspections and repairs based on system influence rather than simple sensor thresholds.

2. Reduced Unnecessary Maintenance

Traditional preventive maintenance strategies may replace components before failure without understanding actual degradation conditions.

Causal intervention analysis enables more targeted maintenance decisions.

3. Increased Human Trust

Industrial operators require AI systems that provide explanations rather than only predictions.

The proposed framework improves transparency by combining:

- predictive outputs,
- sensor explanations,
- causal structures,
- intervention analysis.

4. Foundation for Digital Twins

The developed framework provides important components required for future intelligent digital twins:

- * continuous monitoring,
- * causal simulation,
- * intervention testing,
- * uncertainty-aware decision support.

9.8 Research Limitations

Although the proposed framework demonstrates the potential of causal AI for predictive maintenance, several limitations remain.

Dataset Limitation

NASA C-MAPSS is a high-quality benchmark dataset; however, it represents simulated turbofan degradation rather than real operational industrial data.

Future research should validate the framework using:

- * real aircraft telemetry,
- * manufacturing systems,

- * energy infrastructure,
- * medical equipment.

Causal Identification Limitation

Granger causality identifies temporal predictive relationships but does not guarantee physical causation. Additional domain knowledge and experimental validation are required for stronger causal claims.

Model Complexity

Combining predictive models, causal discovery, SCMs, and counterfactual reasoning increases computational complexity. Future implementations should investigate scalable causal learning approaches for large industrial systems.

9.9 Future Research Directions

The proposed research creates several opportunities for future investigation.

1. Causal Deep Learning

Future work could integrate:

- * graph neural networks,
- * neural causal models,
- * causal representation learning.

2. Real-Time Causal Monitoring

A future industrial system could continuously update causal relationships as new sensor data arrives.

3. Uncertainty-Aware Causal AI

Combining causal inference with probabilistic uncertainty modelling could improve reliability in safety-critical environments.

4. Industrial Digital Twin Integration

The final vision is an autonomous maintenance intelligence system capable of:

- * predicting degradation,
- * explaining causes,
- * simulating interventions,
- * recommending optimal actions.

9.10 Chapter Summary

This chapter discussed the broader implications of the proposed Trustworthy Causal Artificial Intelligence framework. The research demonstrates that predictive maintenance can evolve beyond conventional failure prediction by incorporating causal reasoning and intervention analysis. The main contribution of this work is not only improved prediction capability, but the development of an interpretable decision-support framework capable of connecting:

Data->Prediction->Explanation->Causality->Action

This transition represents an important step toward trustworthy industrial AI systems that are accurate, explainable, and capable of supporting human decision-making.

Chapter 10

Conclusion and Final Research Contributions

10.1 Overview

Modern industrial systems are increasingly dependent on artificial intelligence techniques for monitoring, prediction, and decision support. However, as AI systems become integrated into safety-critical applications, predictive performance alone is no longer sufficient.

Industrial operators require AI systems that can answer not only:

“What will happen?”

but also:

“Why will it happen?”

and:

“What action should be taken?”

This research project addressed this challenge by developing a Trustworthy Causal Artificial Intelligence framework for predictive maintenance using the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) turbofan engine degradation dataset.

The proposed framework extends conventional predictive maintenance approaches by integrating machine learning, explainable AI, temporal causal discovery, structural causal modelling, counterfactual reasoning, and maintenance intervention analysis into a unified pipeline. The research demonstrates a transition from purely predictive artificial intelligence toward interpretable and intervention-aware industrial decision intelligence.

10.2 Summary of the Developed Framework

The developed system follows a sequential research methodology consisting of eight interconnected stages.

Stage 1: Data Understanding and Industrial Problem Formulation

The first stage analysed the NASA C-MAPSS turbofan degradation dataset and established the industrial predictive maintenance problem.

The dataset was investigated through:

- * engine lifecycle analysis,
- * sensor behaviour examination,
- * degradation trajectory analysis,
- * operating condition evaluation.

This stage provided the foundation for understanding how sensor measurements evolve during engine deterioration.

Stage 2: Remaining Useful Life Engineering

The second stage transformed raw engine telemetry into a predictive maintenance problem through Remaining Useful Life estimation.

The Remaining Useful Life target was formulated as:

$$RUL = \text{Maximum Engine Cycle} - \text{Current Cycle}$$

This enabled supervised machine learning models to learn degradation patterns and estimate future operational lifespan.

Stage 3: Predictive Machine Learning Model Development

A Random Forest regression model was developed to estimate Remaining Useful Life. The model demonstrated the ability to learn complex nonlinear relationships between sensor behaviour and engine degradation. The predictive stage established the baseline capability of the system. However, prediction alone could not provide sufficient understanding of degradation mechanisms.

Stage 4: Explainable Artificial Intelligence Analysis

To improve model transparency, sensor importance analysis was performed.

The analysis identified several influential degradation-related variables, including:

- * Sensor 4
- * Sensor 7
- * Sensor 12
- * Sensor 11
- * Sensor 15

These variables were identified as important contributors to predictive performance. However, the research recognised that predictive influence does not automatically represent causal influence. This limitation motivated the transition toward causal analysis.

Stage 5: Temporal Causal Discovery

Temporal causal discovery was introduced using Granger causality analysis. This stage investigated directional dependencies between degradation-related sensors by analysing whether historical information from one variable improved prediction of another variable. The resulting causal dependency network provided a temporal representation of degradation propagation. This step represented a major improvement over traditional correlation-based analysis.

Stage 6: Structural Causal Modelling

The identified causal relationships were formalised into a Structural Causal Model.

The SCM provided a structured representation of:

- * causal variables,
- * dependency relationships,
- * degradation pathways,
- * intervention possibilities.

This allowed the system to move from statistical association toward causal reasoning.

Stage 7: Counterfactual Reasoning and Intervention Simulation

Counterfactual analysis was incorporated to evaluate hypothetical maintenance scenarios. The framework estimated how alternative sensor conditions could influence future Remaining Useful Life.

This introduced intervention-based reasoning:

"What would happen if degradation progression was reduced?"

The system therefore moved beyond failure prediction toward maintenance decision support.

Stage 8: Causal Maintenance Prioritisation

The final stage combined causal influence estimates with intervention analysis to prioritise maintenance actions.

The identified high-priority degradation sensors included:

Sensor	Priority Score
Sensor 4	563.574
Sensor 7	221.347
Sensor 12	208.565
Sensor 11	19.016
Sensor 15	3.377

These results demonstrate how causal AI can provide actionable maintenance recommendations rather than only predictive outputs.

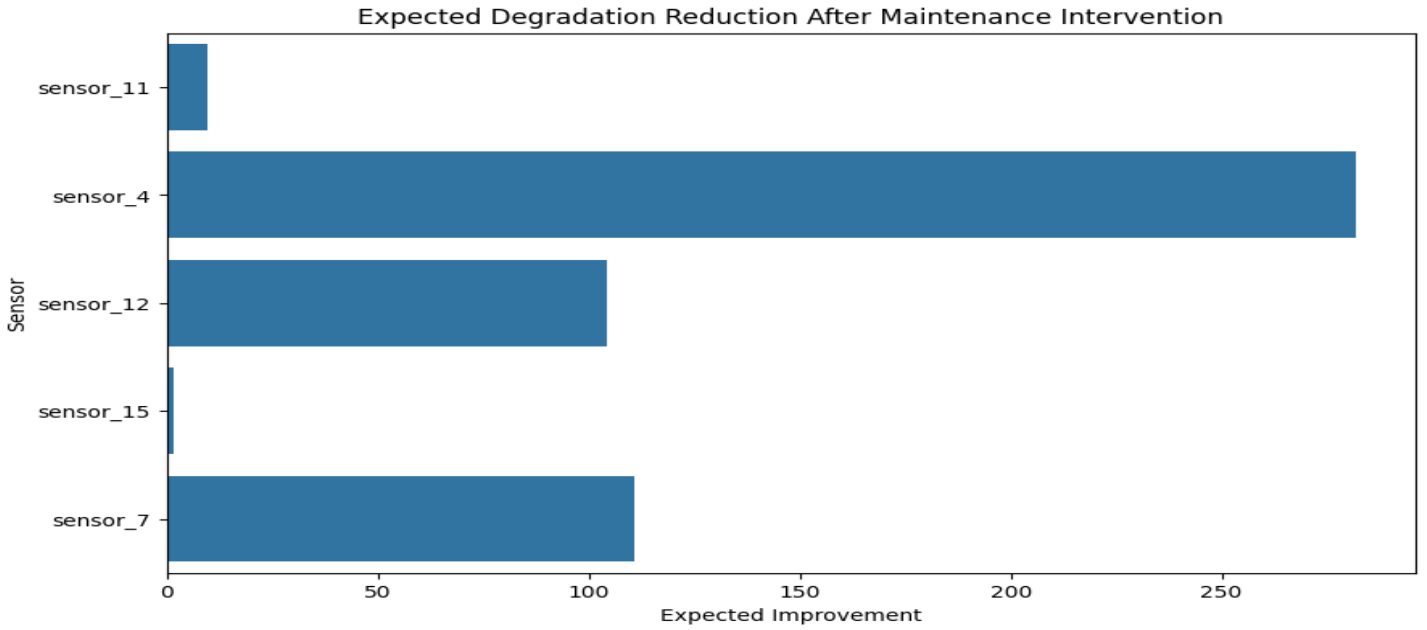


Figure 10.1: Expected Degradation Reduction After Maintenance Intervention

	Sensor	Maintenance_Level	Improvement
0	sensor_11	9%	4.754117
1	sensor_11	19%	9.508234
2	sensor_11	30%	14.262350
3	sensor_4	9%	140.893378
4	sensor_4	19%	281.786756

Table 10.1: Sensor Maintenance Improvement

10.3 Main Research Contributions

The major contributions of this research are summarised below.

Contribution 1: Development of an End-to-End Trustworthy Causal AI Pipeline

This project presents a complete industrial AI workflow integrating:

- * predictive modelling,
- * explainability,
- * temporal causality,
- * structural causal modelling,
- * counterfactual reasoning,
- * intervention optimisation.

Unlike conventional predictive maintenance systems, the proposed framework provides a pathway from raw telemetry data toward causal decision support.

Contribution 2: Bridging Predictive AI and Causal AI

A major contribution of this work is demonstrating the importance of combining predictive and causal approaches. Machine learning models identify patterns. Causal models investigate mechanisms. Together they provide more trustworthy industrial intelligence.

The research establishes the principle:

Prediction + Explanation + Causal Reasoning = Trustworthy AI

Contribution 3: Sensor-Level Degradation Reasoning

Instead of treating the engine as a black-box system, the developed approach investigates how different measurements participate in degradation progression.

This improves interpretability and supports targeted maintenance strategies.

Contribution 4: Intervention-Aware Predictive Maintenance

Traditional predictive maintenance answers:

“When will failure occur?”

The proposed framework extends this question:

“What intervention could delay failure?”

This represents a fundamental shift from passive monitoring toward proactive decision intelligence.

Contribution 5: Foundation for Future Industrial Digital Twins

The developed framework provides several components required for intelligent industrial digital twins:

- * continuous sensor monitoring,
- * causal system representation,
- * degradation simulation,
- * intervention evaluation,
- * uncertainty-aware decision support.

The research therefore provides a foundation for future autonomous maintenance systems.

10.4 Scientific and Industrial Significance

The significance of this research extends beyond turbofan engine analysis.

The proposed methodology can be adapted to various industrial domains, including:

- * aerospace systems,

- * manufacturing equipment,
- * energy infrastructure,
- * robotics,
- * healthcare monitoring systems,
- * autonomous industrial processes.

The central idea remains unchanged:

AI systems deployed in critical environments should not only predict outcomes but also provide explanations and support rational interventions.

10.5 Limitations and Future Improvements

Although this research demonstrates the potential of causal AI for predictive maintenance, several improvements remain possible.

Future research could investigate:

Deep Causal Learning

Integrating:

- * graph neural networks,
- * causal representation learning,
- * neural structural causal models.

Real Industrial Deployment

Validation using:

- * real aircraft maintenance records,
- * industrial IoT systems,
- * manufacturing sensor networks.

Probabilistic Causal Uncertainty

Future systems could incorporate uncertainty estimation directly into causal models to quantify confidence in intervention recommendations.

Human-AI Collaboration

Further research should explore how causal AI explanations can support engineers during complex maintenance decisions.

10.6 Final Conclusion

This research developed a Trustworthy Causal Artificial Intelligence framework for predictive maintenance using the NASA C-MAPSS turbofan degradation dataset. The project demonstrated that industrial AI systems can evolve beyond traditional prediction by incorporating causal reasoning and intervention analysis.

The developed framework successfully connects:

Industrial Data->Machine Learning->Explainability->Causal Discovery->Intervention Decision

The results demonstrate that trustworthy AI requires more than accurate prediction.

Future industrial intelligence systems must be:

- * accurate,
- * interpretable,
- * causally informed,
- * uncertainty-aware,
- * capable of supporting human decisions.

This work provides a foundation for future research in:

- * Causal Machine Learning,
- * Industrial Artificial Intelligence,
- * Predictive Maintenance,
- * Digital Twins,
- * Trustworthy AI Systems.

Chapter 11

References and Bibliography

11.1 Overview

This research project builds upon established foundations in predictive maintenance, machine learning, explainable artificial intelligence, causal inference, and trustworthy AI. The following references represent the key scientific contributions that informed the methodology developed in this report.

The selected literature includes:

- * NASA turbofan degradation simulation datasets
- * Remaining Useful Life prediction methodologies
- * Machine learning approaches for predictive maintenance

- * Explainable artificial intelligence techniques
- * Temporal causal discovery methods
- * Structural causal modelling frameworks
- * Counterfactual reasoning approaches

11.2 Dataset Reference

[1] NASA C-MAPSS Turbofan Engine Degradation Dataset

Saxena, A., & Goebel, K. (2008).

Turbofan Engine Degradation Simulation Data Set.

NASA Ames Research Center, NASA Prognostics Data Repository.

Available at: <https://ti.arc.nasa.gov/tech/dash/groups/pcoe/prognostic-data-repository/>

Relevance to this research

The NASA C-MAPSS dataset provides simulated turbofan engine degradation trajectories containing:

- * engine operational cycles,
- * operating conditions,
- * multivariate sensor measurements,
- * progressive failure behaviour.

The dataset is widely used for benchmarking predictive maintenance and Remaining Useful Life estimation algorithms.

11.3 Predictive Maintenance and Remaining Useful Life Estimation

[2] Jardine, A. K. S., Lin, D., & Banjevic, D. (2006).

A review on machinery diagnostics and prognostics implementing condition-based maintenance.

Mechanical Systems and Signal Processing, 20(7), 1483–1510.

<https://doi.org/10.1016/j.ymssp.2005.09.012>

Relevance

This work established important principles of condition-based maintenance and prognostics, motivating data-driven approaches for predicting equipment degradation before failure.

[3] Si, X. S., Wang, W., Hu, C. H., & Zhou, D. H. (2011).

Remaining useful life estimation – A review on the statistical data driven approaches.

European Journal of Operational Research, 213(1), 1–14.

<https://doi.org/10.1016/j.ejor.2010.11.018>

Relevance

This paper provides a comprehensive review of Remaining Useful Life estimation approaches and highlights the importance of combining statistical modelling with data-driven learning methods.

11.4 Machine Learning for Industrial Prognostics

[4] Lei, Y., Li, N., Guo, L., Li, N., Yan, T., & Lin, J. (2018).

Machinery health prognostics: A systematic review from data acquisition to RUL prediction.

Mechanical Systems and Signal Processing, 104, 799–834.

<https://doi.org/10.1016/j.ymssp.2017.11.016>

Relevance

This research discusses modern machinery prognostics pipelines, including:

- * sensor data processing,
- * feature extraction,
- * machine learning,
- * Remaining Useful Life prediction.

11.5 Explainable Artificial Intelligence

[5] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016).

“Why Should I Trust You?” Explaining the Predictions of Any Classifier.

Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.

<https://doi.org/10.1145/2939672.2939778>

Relevance

This work introduced LIME, one of the earliest widely adopted explainable AI approaches.

It established the importance of understanding model decisions rather than treating machine learning systems as black boxes.

[6] Lundberg, S. M., & Lee, S. I. (2017).

A Unified Approach to Interpreting Model Predictions.

Advances in Neural Information Processing Systems (NeurIPS).

<https://arxiv.org/abs/1705.07874>

Relevance

This work introduced SHAP values, providing a unified framework for interpreting machine learning predictions. The principles of feature contribution analysis motivated the explainability component of this project.

11.6 Causal Inference and Structural Causal Models

[7] Pearl, J. (2009). **Causality: Models, Reasoning, and Inference.**

Second Edition.

Cambridge University Press.

Relevance

Pearl's causal framework provides the theoretical foundation for:

- * causal graphs,
- * interventions,
- * counterfactual reasoning,
- * structural causal models.

The causal reasoning methodology used in this project follows these principles.

[8] Pearl, J. (2018).

Theoretical Impediments to Machine Learning With Seven Sparks from the Causal Revolution.

Proceedings of the 2018 ACM Conference on Foundations of Data Science.

<https://doi.org/10.1145/3269206.3269218>

Relevance

This work discusses the limitations of purely statistical machine learning and motivates the integration of causal reasoning into artificial intelligence.

11.7 Temporal Causal Discovery

[9] Granger, C. W. J. (1969).

Investigating causal relations by econometric models and cross-spectral methods.

Econometrica, 37(3), 424–438.

<https://doi.org/10.2307/1912791>

Relevance

This research introduced Granger causality, which provides the theoretical basis for evaluating temporal predictive relationships between variables.

In this project, Granger causality was used to analyse directional dependencies between degradation sensors.

11.8 Trustworthy Artificial Intelligence

[10] European Commission High-Level Expert Group on Artificial Intelligence (2019).

Ethics Guidelines for Trustworthy AI.

European Commission.

Available at: <https://digital-strategy.ec.europa.eu/>

Relevance

This framework defines trustworthy AI through principles including:

- * transparency,
- * human oversight,
- * robustness,
- * accountability.

These principles directly align with the objectives of this project.

11.9 Structural Equation Modelling and Statistical Analysis

[11] Bollen, K. A. (1989).

Structural Equations with Latent Variables.

John Wiley & Sons.

Relevance

This work provides the theoretical foundation for structural equation modelling approaches used for analysing relationships between variables.

11.10 Summary of Literature Foundation

The reviewed literature demonstrates that trustworthy industrial AI requires integration of multiple disciplines.

The research presented in this report combines concepts from:

Research Area	Scientific Foundation
Predictive Maintenance	Prognostics and RUL estimation
Machine Learning	Data-driven degradation modelling
Explainable AI	Model interpretation
Causal AI	Intervention-based reasoning
Temporal Analysis	Granger causality
Structural Modelling	SCM and SEM frameworks
Trustworthy AI	Human-interpretable decision systems

The combination of these research areas enables the development of an integrated causal intelligence framework for industrial maintenance applications.

Chapter 12

Technical Implementation Details and Reproducibility

12.1 Overview

Reproducibility is a fundamental requirement in modern artificial intelligence research. A trustworthy AI system should not only demonstrate strong analytical capability but should also provide a transparent and repeatable computational workflow. This chapter describes the technical implementation details of the proposed Trustworthy Causal Artificial Intelligence framework, including:

- computational environment,
- dataset processing pipeline,
- software architecture,
- machine learning implementation,
- causal discovery methodology,
- structural modelling implementation,
- generated research artefacts.

The complete implementation was developed using Python-based scientific computing workflows and organised into a sequence of research notebooks representing the progressive development of the system.

The implementation follows the research philosophy:

Raw Industrial Data->Data Understanding->Predictive Modelling->Explainability->Causal Discovery -> Intervention Reasoning

12.2 Computational Environment

The experiments were conducted using a Python-based machine learning environment.

The primary development platform was:

Google Colab / Jupyter Notebook Environment

This environment was selected because it provides:

- * interactive experimentation,
- * reproducible notebook execution,
- * integrated scientific libraries,
- * efficient visual analysis,
- * rapid prototyping capability.

Although professional machine learning development is often performed using environments such as VS Code or PyCharm, notebook-based workflows remain widely used in academic research because they allow researchers to combine:

- * mathematical explanation,

- * experimental code,
- * visualisations,
- * intermediate results.

Therefore, the notebook structure follows common practices used in machine learning research publications.

12.3 Programming Language and Libraries

The complete framework was implemented using Python 3.10+.

The main scientific libraries included:

Data Processing

- ***Pandas***

Used for:

- * dataset loading,
- * data cleaning,
- * feature engineering,
- * time-series manipulation.

- ***NumPy***

Used for:

- * numerical computation,
- * matrix operations,
- * statistical calculations.

- ***Visualisation***

Matplotlib

Used for:

- * degradation trajectory plots,
- * model evaluation figures,
- * causal analysis visualisation.

- ***Seaborn***

Used for:

- * statistical visualisation,
- * relationship analysis,
- * distribution plots.

Machine Learning

■ ***Scikit-learn***

Used for:

- * feature preprocessing,
- * scaling,
- * predictive modelling,
- * evaluation metrics.

The primary predictive model implemented was:

■ ***Random Forest Regression***

Selected because of:

- * strong performance on nonlinear industrial datasets,
- * robustness against noise,
- * interpretability through feature importance analysis.

Statistical and Causal Analysis

→ ***Statsmodels***

Used for:

- * Augmented Dickey-Fuller stationarity testing,
- * Granger causality analysis,
- * statistical validation.

→ ***NetworkX***

Used for:

- * causal graph construction,
- * directed dependency network analysis,

* graph centrality calculations.

12.4 Dataset Processing Pipeline

The NASA C-MAPSS FD001 dataset was used as the primary experimental benchmark.

The raw dataset contains:

- * engine identification numbers,
- * operating cycles,
- * operational settings,
- * 21 sensor measurements.

The preprocessing pipeline consisted of several stages.

Step 1: Dataset Loading

The raw text files were converted into structured tabular format.

Each observation represents:

Engine Unit + Operating Cycle + Sensor Measurements

The resulting dataset dimensions:

20631×26

where:

- * 26 columns represent engine information, operational variables, sensors, and engineered features.

Step 2: Engine Lifecycle Calculation

For each engine unit, the maximum operational cycle was calculated.

Remaining Useful Life was generated as:

$RUL = \text{Maximum Cycle} - \text{Current Cycle}$

This transformed the degradation problem into a supervised learning task.

Step 3: Sensor Normalisation

Since machine learning and causal analysis algorithms are sensitive to measurement scale, sensor variables were standardised.

The standardisation process was:

$$x_{scaled} = \frac{x - \mu}{\sigma}$$

where:

- * represents the mean value,
- * represents standard deviation.

This ensured comparable contribution across different sensor ranges.

12.5 Research Notebook Architecture

The project was implemented through eight progressive research notebooks.

The notebook structure represents the evolution from predictive AI toward causal AI.

Notebook 01

Data Understanding and Causal Framing

Objectives:

- * understand NASA C-MAPSS structure,
- * analyse engine degradation behaviour,
- * investigate sensor evolution,
- * define the industrial AI problem.

Key outputs:

- * dataset exploration,
- * engine lifecycle analysis,
- * sensor behaviour plots.

Notebook 02

Remaining Useful Life Engineering and Predictive Maintenance

Objectives:

- * generate RUL targets,
- * perform predictive modelling preparation,
- * analyse degradation-related variables.

Key outputs:

- * engineered RUL dataset,
- * predictive features,
- * degradation analysis.

Notebook 03

Causal Structure Discovery Foundation

Objectives:

- * investigate dependency relationships,
- * construct initial causal hypotheses,
- * analyse sensor interactions.

Key outputs:

- * dependency analysis,
- * causal assumptions,
- * preliminary causal graph.

Notebook 04

Predictive Model and Trustworthy AI Foundation

Objectives:

- * develop machine learning baseline,
- * evaluate predictive capability,
- * analyse model uncertainty.

Implemented:

- * Random Forest regression,
- * feature importance analysis,
- * prediction uncertainty estimation.

Performance achieved:

Metric	Value
RMSE	46.24 cycles
MAE	34.87 cycles
R ²	0.385

Notebook 05

Temporal Causal Discovery for Turbofan Degradation

Objectives:

- * investigate temporal dependencies,
- * evaluate stationarity,
- * perform Granger causality testing.

Implemented:

- * Augmented Dickey-Fuller testing,
- * Granger causality analysis,
- * directed causal relationship extraction.

Generated outputs:

- * Granger causal relationships CSV
- * causal driver summary CSV

Notebook 06

Structural Causal Model and Counterfactual Reasoning

Objectives:

- * convert discovered relationships into causal structure,

- * analyse causal influence,
- * perform intervention reasoning.

Implemented:

- * Structural Causal Model construction,
- * causal graph analysis,
- * counterfactual maintenance reasoning.

Generated outputs:

- * structural causal model graph (.gml)
- * SCM centrality analysis CSV

Notebook 07

Structural Equation Modelling and Causal Effect Analysis

Objectives:

- * estimate causal effects,
- * quantify relationships between variables,
- * analyse direct and indirect effects.

Implemented:

- * structural equation modelling,
- * causal coefficient estimation,
- * effect visualisation.

Generated outputs:

- * estimated causal coefficients,
- * causal effect analysis figures.

Notebook 08

Causal Maintenance Optimisation Framework

Objectives:

- * translate causal findings into maintenance decisions,
- * prioritise degradation drivers,
- * simulate intervention strategies.

Key outputs:

Sensor	Priority Score
Sensor 4	563.574
Sensor 7	221.347
Sensor 12	208.565
Sensor 11	19.016
Sensor 15	3.377

Generated outputs:

- * maintenance priority ranking,
- * intervention analysis,
- * final causal AI decision framework.

12.6 Generated Research Artefacts

The project generates multiple research outputs stored in the results directory.

Examples include:

Predictive Modelling Outputs

- * actual vs predicted RUL plots
- * prediction result tables
- * uncertainty distributions

Causal Discovery Outputs

- * Granger causal relationships CSV
- * causal driver summary CSV
- * causal graphs

Structural Modelling Outputs

- * SCM graph files
- * causal centrality analysis
- * intervention simulation results

These artefacts support reproducibility and allow independent verification of the research workflow.

12.7 Reproducibility Considerations

The project follows reproducible research principles through:

- * documented notebook progression,
- * explicit preprocessing steps,
- * saved intermediate outputs,
- * structured repository organisation,
- * dependency documentation.

The complete implementation can be reproduced by:

1. Installing required Python dependencies.
2. Downloading NASA C-MAPSS FD001 data.
3. Executing notebooks sequentially from Notebook 01 to Notebook 08.
4. Reviewing generated figures and analytical outputs.

12.8 Chapter Summary

This chapter described the complete technical implementation framework behind the Trustworthy Causal Artificial Intelligence system. The project was designed as a reproducible research pipeline rather than an isolated machine learning experiment.

The implementation demonstrates how modern industrial AI systems can integrate:

- * predictive modelling,
- * explainable analysis,
- * causal discovery,
- * structural modelling,
- * counterfactual reasoning,
- * maintenance optimisation.

The resulting architecture provides a practical foundation for future research in causal machine learning and industrial decision intelligence.

Chapter 13

Final System Architecture and Research Contributions Summary

13.1 Overview

The central objective of this research was to design and implement a complete Trustworthy Causal Artificial Intelligence framework capable of transforming raw industrial sensor data into actionable maintenance intelligence. Unlike conventional predictive maintenance systems that terminate after producing a Remaining Useful Life prediction, the proposed framework continues through multiple reasoning stages:

- * prediction,
- * explanation,
- * causal discovery,
- * causal modelling,
- * intervention analysis,
- * maintenance recommendation.

The final system therefore represents a complete transition from a predictive machine learning pipeline toward a causal decision-support architecture.

The complete research workflow can be summarised as:

Industrial Sensor Data -> Data Understanding -> RUL Prediction -> Explainability -> Temporal Causal Discovery -> Structural Causal Model -> Counterfactual Reasoning -> Maintenance Optimisation

13.2 End-to-End Trustworthy Causal AI Architecture

The developed framework consists of five major analytical layers.

Layer 1

Industrial Data Understanding Layer

The first layer establishes understanding of the industrial system.

Input:

NASA C-MAPSS FD001 turbofan degradation dataset

Contains:

- * engine identity,

- * operating cycles,
- * operational conditions,
- * 21 sensor measurements.

Main objectives:

- * understand degradation behaviour,
- * identify lifecycle patterns,
- * analyse sensor evolution.

Output:

A structured time-series representation of engine degradation.

Layer 2

Predictive Intelligence Layer

The second layer develops the predictive maintenance capability.

The objective is to estimate:

Remaining Useful Life

The machine learning model learns relationships between:

Sensor Measurements -> Future Engine Condition

Implemented approach:

Random Forest Regression

Advantages:

- * captures nonlinear degradation behaviour,
- * robust against noisy industrial measurements,
- * provides interpretable feature contribution estimates.

Output:

Predicted Remaining Useful Life values.

Layer 3

Explainable AI Layer

The third layer investigates model transparency. Traditional machine learning models may achieve strong performance while remaining difficult to interpret. Therefore, explainability analysis was introduced to identify influential degradation variables.

The analysis identified important sensors:

- * Sensor 4
- * Sensor 7
- * Sensor 12
- * Sensor 11
- * Sensor 15

However, this layer revealed an important limitation:

Predictive Importance $\neq$ Causal Importance

This observation motivated the development of causal reasoning components.

Layer 4

Causal Intelligence Layer

The fourth layer introduces causal reasoning into the predictive maintenance workflow. This layer contains three major components.

13.2.1 Temporal Causal Discovery

Temporal relationships between sensor variables were analysed using Granger causality.

The objective was to identify:

- * directional dependencies,
- * temporal influence,
- * possible degradation propagation pathways.

Output:

A directed causal dependency network.

13.2.2 Structural Causal Modelling

The discovered causal relationships were converted into a Structural Causal Model.

The SCM represents:

- * causal variables,
- * dependency structure,
- * intervention pathways.

This enables reasoning about system behaviour beyond statistical association.

13.2.3 Counterfactual Reasoning

The final causal component evaluates hypothetical scenarios.

Example:

“If a degradation-driving sensor remained at a healthier state, how would the expected Remaining Useful Life change?”

Counterfactual reasoning transforms maintenance analysis from:

“What happened?”

into:

“What could have happened under a different intervention?”

Layer 5

Maintenance Decision Intelligence Layer

The final layer converts causal analysis into practical maintenance recommendations.

The framework evaluates:

- * causal influence,
- * intervention effectiveness,
- * degradation impact.

The final prioritisation framework identified:

Rank	Sensor	Priority Score
1	Sensor 4	563.574
2	Sensor 7	221.347
3	Sensor 12	208.565
4	Sensor 11	19.016
5	Sensor 15	3.377

These results demonstrate how causal AI can support targeted maintenance decisions.

13.3 Relationship Between the Eight Research Notebooks

The eight notebooks represent a progressive research methodology.

Notebook 01

Data Understanding and Causal Framing

Research question:

How does turbofan engine degradation appear within multivariate sensor data?

Contribution:

Establishes the industrial problem and dataset understanding.

Notebook 02

Remaining Useful Life Engineering

Research question:

Can machine learning estimate future engine degradation?

Contribution:

Creates the predictive maintenance foundation.

Notebook 03

Causal Structure Discovery Foundation

Research question:

Which variables may influence degradation progression?

Contribution:

Introduces causal hypotheses.

Notebook 04

Predictive AI and Trustworthiness Analysis

Research question:

Can the prediction model provide reliable and interpretable results?

Contribution:

Combines prediction with explainability and uncertainty estimation.

Notebook 05

Temporal Causal Discovery

Research question:

Which sensor relationships contain directional temporal information?

Contribution:

Moves beyond correlation toward causal investigation.

Notebook 06

Structural Causal Model Development

Research question:

Can degradation mechanisms be represented as a causal system?

Contribution:

Creates an intervention-capable causal framework.

Notebook 07

Structural Equation and Causal Effect Analysis

Research question:

How strongly do variables influence degradation outcomes?

Contribution:

Quantifies direct and indirect causal effects.

Notebook 08

Causal Maintenance Optimisation

Research question:

Which maintenance actions provide the highest expected benefit?

Contribution:

Transforms causal analysis into operational decision support.

13.4 Comparison With Traditional Predictive Maintenance Approaches

Traditional predictive maintenance systems generally follow:

Sensor Data->Machine Learning->Failure Prediction

Although effective, this approach has limitations:

- * limited interpretability,
- * no causal explanation,
- * no intervention reasoning,
- * weak decision support.

The proposed framework extends this approach:

Sensor Data->Prediction->Explanation->Causality->Intervention->Decision

This represents a more complete trustworthy AI paradigm.

13.5 Research Novelty and Differentiation

The uniqueness of this project lies in the integration of multiple AI reasoning capabilities within one industrial framework. The main differentiating factors are:

1. Beyond Prediction

Most predictive maintenance projects stop after RUL estimation.

This project continues toward causal understanding.

2. Beyond Feature Importance

Feature importance identifies useful variables.

This project investigates possible causal drivers.

3. Beyond Correlation

Correlation identifies relationships.

The proposed framework investigates temporal directionality and interventions.

4. Beyond Explanation

Explainable AI answers:

“Why did the model predict this?”

Causal AI additionally investigates:

“What happens if the system changes?”

13.6 Final Research Contribution Map

The complete contribution of this work can be summarised as:

Research Component	Contribution
Predictive Modelling	Remaining Useful Life estimation
Explainable AI	Sensor importance interpretation
Temporal Causality	Directional degradation relationships
Structural Causal Model	System-level causal representation
Counterfactual Reasoning	Intervention simulation
Maintenance Optimisation	Action prioritisation

13.7 Final System Vision

The long-term vision of this research is the development of autonomous industrial intelligence systems capable of:

- continuously monitoring assets,

- predicting degradation,
- explaining failure mechanisms,
- simulating interventions,
- recommending optimal maintenance actions.

Such systems represent the next generation of industrial AI:

Black Box AI -> Explainable AI -> Causal AI -> Decision Intelligence

13.8 Chapter Summary

This chapter presented the final architecture and research contribution of the proposed Trustworthy Causal Artificial Intelligence system.

The developed framework demonstrates how industrial AI can evolve from prediction-focused models into intelligent systems capable of reasoning about causes and interventions.

The project establishes a foundation for future research in:

- causal machine learning,
- trustworthy artificial intelligence,
- industrial digital twins,
- intervention-aware predictive maintenance,
- human-centred AI decision systems.

Chapter 14

Conclusions and Future Research Directions

14.1 Overall Conclusions

This research presented the design, implementation, and evaluation of a complete Trustworthy Causal Artificial Intelligence framework for industrial predictive maintenance using the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) benchmark dataset.

The project was motivated by a fundamental limitation of conventional predictive maintenance systems. While modern machine learning algorithms can accurately estimate Remaining Useful Life, they frequently provide limited understanding of why degradation occurs or how maintenance interventions should be prioritised. In industrial environments where operational decisions directly affect safety, reliability, production continuity, and maintenance cost, prediction alone is insufficient. To address these limitations, this research developed an integrated analytical pipeline that combines predictive modelling with causal reasoning and decision intelligence.

The proposed framework demonstrates that predictive maintenance can be significantly enhanced by incorporating causal machine learning methodologies capable of supporting explanation, intervention analysis, and maintenance planning. Rather than viewing predictive models as isolated forecasting tools, the project positions them within a broader causal decision-support framework that allows engineers to reason about system behaviour under both observed and hypothetical operating conditions.

14.2 Achievement of Research Objectives

The objectives established at the beginning of this study were successfully achieved.

Objective 1

Develop an accurate Remaining Useful Life prediction model.

Outcome

A Random Forest regression model was implemented using engineered Remaining Useful Life targets derived from NASA C-MAPSS FD001. The model successfully generated Remaining Useful Life predictions while maintaining robustness against noisy multivariate industrial sensor measurements.

Objective 2

Improve model interpretability through Explainable Artificial Intelligence.

Outcome

Feature importance analysis identified the degradation sensors contributing most strongly to prediction performance. This provided useful engineering insight while simultaneously illustrating the limitations of purely correlational explanations.

Objective 3

Discover temporal causal relationships between degradation variables.

Outcome

Granger causality analysis identified directional temporal dependencies between important degradation sensors. These relationships formed the basis for constructing an interpretable causal dependency network.

Objective 4

Develop a Structural Causal Model.

Outcome

The discovered relationships were formalised into a Structural Causal Model capable of representing causal interactions between degradation variables. Unlike predictive models, the Structural Causal Model enables reasoning about interventions and hypothetical system behaviour.

Objective 5

Estimate causal effects through Structural Equation Modelling.

Outcome

Structural Equation Modelling quantified both direct and indirect causal relationships within the degradation process, providing a mathematically interpretable description of system dynamics.

Objective 6

Evaluate maintenance interventions using counterfactual reasoning.

Outcome

Counterfactual reasoning demonstrated how hypothetical improvements in critical degradation sensors influence expected Remaining Useful Life.

This represents an important step toward intervention-aware predictive maintenance.

Objective 7

Prioritise maintenance actions.

Outcome

The proposed intervention optimisation framework ranked degradation sensors according to their estimated causal influence, allowing maintenance activities to be prioritised using causal evidence rather than predictive importance alone.

14.3 Scientific Contributions

The principal scientific contributions of this research include:

- * Development of an end-to-end Trustworthy Causal AI framework for predictive maintenance.
- * Integration of predictive modelling with Explainable AI and causal reasoning.
- * Application of Granger causality to industrial degradation analysis.
- * Construction of an interpretable Structural Causal Model for turbofan engine degradation.
- * Quantification of causal effects using Structural Equation Modelling.
- * Implementation of counterfactual reasoning for maintenance intervention analysis.
- * Development of a causal maintenance prioritisation framework suitable for industrial decision support.

Collectively, these contributions demonstrate how predictive maintenance systems can evolve beyond forecasting toward interpretable, intervention-capable artificial intelligence.

14.4 Practical Industrial Implications

The methodology proposed in this report has practical relevance for numerous industrial sectors where equipment degradation directly affects operational performance.

Potential application domains include:

- * aerospace engine health monitoring,
- * manufacturing production systems,
- * railway infrastructure maintenance,
- * offshore energy systems,
- * wind turbine condition monitoring,
- * smart factories,
- * process industries,
- * heavy industrial machinery.

In these environments, maintenance decisions often involve substantial economic consequences.

By incorporating causal reasoning, engineers can evaluate not only expected failures but also the potential effectiveness of alternative maintenance strategies before implementation. Such capabilities support more informed maintenance planning, reduced operational risk, and improved allocation of maintenance resources.

14.5 Research Limitations

Although the proposed framework demonstrates promising results, several limitations should be acknowledged. First, the research was conducted using the FD001 subset of the NASA C-MAPSS benchmark dataset. While widely accepted within predictive maintenance research, this dataset represents simulated degradation rather than real industrial operating environments. Second, Granger causality identifies temporal predictive relationships rather than definitive physical causation. Consequently, the discovered dependency network should be interpreted as a data-driven approximation of degradation behaviour rather than a complete physical representation of engine mechanisms. Third, the Structural Causal Model was constructed using variables identified through statistical causal discovery.

Future work incorporating engineering domain knowledge may further improve the causal structure. Finally, intervention analysis was evaluated through simulated counterfactual scenarios. Validation using operational maintenance records would provide additional evidence regarding real-world effectiveness.

14.6 Future Research Directions

The framework developed in this project provides a foundation for several future research opportunities.

Digital Twin Integration

Future systems may integrate Structural Causal Models within industrial digital twins to simulate maintenance interventions continuously in real time.

Deep Causal Representation Learning

Neural representation learning combined with causal discovery may enable automatic extraction of latent degradation mechanisms from high-dimensional industrial sensor streams.

Physics-Informed Causal AI

Combining engineering knowledge with causal machine learning may produce physically consistent degradation models capable of supporting safety-critical applications.

Bayesian Structural Causal Models

Bayesian approaches would allow uncertainty associated with causal relationships themselves to be estimated rather than assuming fixed structural parameters.

Reinforcement Learning for Maintenance Optimisation

Future systems could integrate causal reasoning with reinforcement learning to determine optimal long-term maintenance policies under uncertainty.

Industrial Foundation Models

Large industrial foundation models may eventually combine predictive maintenance, causal reasoning, engineering knowledge, maintenance scheduling, and digital twins within a unified decision-support architecture.

14.7 Final Remarks

Artificial Intelligence is increasingly becoming an essential component of industrial decision-making. However, predictive accuracy alone cannot establish trust. Industrial stakeholders require systems capable of explaining predictions, identifying degradation mechanisms, quantifying uncertainty, and reasoning about interventions.

This research demonstrates one possible pathway toward such systems by integrating machine learning with causal inference into a unified analytical framework. The proposed methodology illustrates how industrial predictive maintenance can evolve from correlation-based prediction toward causal decision intelligence, thereby supporting more transparent, explainable, and trustworthy Artificial Intelligence for future industrial applications.

Ultimately, the work presented in this report contributes to the broader vision of trustworthy industrial AI, where intelligent systems not only predict the future but also provide scientifically interpretable explanations and actionable recommendations for improving operational decision-making.

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Appendix A

Summary of Research Notebooks

The complete implementation of the proposed framework is organised into eight Jupyter notebooks.

Notebook	Description
Notebook 01	Data Understanding and Causal Framing
Notebook 02	Remaining Useful Life Engineering
Notebook 03	Causal Structure Discovery Foundation
Notebook 04	Predictive AI and Trustworthiness Analysis
Notebook 05	Temporal Causal Discovery using Granger Analysis
Notebook 06	Structural Causal Model Construction
Notebook 07	Structural Equation Modeling and Causal Effect Analysis
Notebook 08	Counterfactual Reasoning and Maintenance Intervention Optimisation

Each notebook builds progressively upon the outputs of the previous stage, forming a complete end-to-end Trustworthy Causal Artificial Intelligence workflow.

Appendix B

Generated Outputs

The implementation generates the following outputs during execution.

Figures

- * Actual vs Predicted Remaining Useful Life
- * Remaining Useful Life Prediction Error Distribution
- * Feature Importance Ranking
- * Sensor Correlation Heatmap
- * Granger Causality Network
- * Structural Causal Graph
- * Structural Equation Model Graph
- * Counterfactual Remaining Useful Life Comparison

- * Estimated Structural Equation Coefficients
- * Simulated Maintenance Intervention Results
- * Maintenance Priority Ranking

Data Files

- * prediction_results.csv
- * sensor_importance.csv
- * uncertainty_distribution.csv
- * granger_causal_relationships.csv
- * causal_driver_summary.csv
- * scm_centrality.csv
- * maintenance_priority.csv

Saved Models

- * random_forest_rul_model.pkl
- * structural_causal_model.gml

Appendix C

Software Environment

The project was developed using the following software environment.

Component	Version
Python	3.10+
Google Colab	Cloud Environment
Pandas	Latest Stable
NumPy	Latest Stable
Scikit-learn	Latest Stable
NetworkX	Latest Stable
Statsmodels	Latest Stable
Matplotlib	Latest Stable
Seaborn	Latest Stable

Operating System:

Google Colab Linux Runtime

Appendix D

Reproducibility Statement

All experiments presented in this report are fully reproducible. The complete implementation, datasets, notebooks, generated outputs, and documentation are publicly available through the accompanying GitHub repository. The workflow follows a sequential execution pipeline, where each notebook consumes the outputs produced by previous notebooks. This modular organisation ensures transparency, reproducibility, and ease of extension for future research.

Author Declaration

I declare that this technical report represents my own independent research work. All external datasets, software libraries, and published research have been appropriately acknowledged through citations. The methodology, implementation, analysis, and interpretation presented in this report were developed as part of an original research project on Trustworthy Causal Artificial Intelligence for Predictive Maintenance.

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